DIVISION OF NAVOTAS CITY

Senior High School

GENERAL MATHEMATICS

(Quarter 1)

First Semester

ADM
ALTERNATIVE DELIVERY MODE

S.Y. 2021-2022
NAVOTAS CITY PHILIPPINES

GOVERNMENT PROPERTY
NOT FOR SALE

General Mathematics for Senior High School
Alternative Delivery Mode
1st Semester (Quarter 1)
Second Edition, 2021

Republic Act 8293, section 176 states that: No copyright shall subsist in any work of the Government of the Philippines. However, prior approval of the government agency or office wherein the work is created shall be necessary for exploitation of such work for profit. Such agency or office may, among other things, impose as a condition the payment of royalties.

Borrowed materials (i.e., songs, stories, poems, pictures, photos, brand names, trademarks, etc.) included in this module are owned by their respective copyright holders. Every effort has been exerted to locate and seek permission to use these materials from their respective copyright owners. The publisher and authors do not represent nor claim ownership over them.

Published by the Department of Education
Secretary: Leonor Magtolis Briones
Undersecretary: Diosdado M. San Antonio

Development Team of the Module

Writer: Jeffrey C. Faustino, Angelo T. Leoncio, and Maureen M. Neri

Editors: Editha O. Peregrino

Reviewers: Alberto J. Tiangco

Illustrator : Jeffrey C. Faustino, Angelo T. Leoncio, and Maureen M. Neri

Layout Artist: Joseph G. Bantug

Management Team: Alejandro G. Ibañez, OIC- Schools Division Superintendent
Isabelle S. Sibayan, OIC- Asst. Schools Division Superintendent
Loida O. Balasa, Chief, Curriculum Implementation Division
Alberto J. Tiangco, EPS in Mathematics
Grace R. Nieves, EPS In Charge of LRMS
Lorena J. Mutas, ADM Coordinator
Vergel Junior C. Eusebio, PDO II LRMS

Inilimbag sa Pilipinas ng _____

Department of Education – Navotas City
Office Address: BES Compound M. Naval St. Sipac-Almacen Navotas City
Telefax: 02-8332-77-64
E-mail Address: Navotas.city@deped.gov.ph

Table of Contents

What I Know 1

Module 1.....3

Module 2.....8

Module 3.....16

Module 4.....25

Module 5.....34

Assessment.....43

Answer Key 44

References 46

What I Know

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

1. This is a rule of correspondence between two nonempty sets, such that each element of the first set called domain, there corresponds one and only one element of the second set called range.

a. Relation b. Function c. Differentiation d. Integration

2. A graph of a mathematical relation is a function if any _____ drawn passing through the graph intersects the graph at exactly one point.

a. Horizontal Line b. Vertical Line c. Diagonal Line d. Both A and B

3.Which of the following sets of ordered pairs are functions?

a. A = {(1, 2), (2, 3), (3, 4), (4, 5)}

b. A = {(1, 1), (2, 2), (3, 3), (4, 4)}

c. A = {(1, 0), (0, 1), (-1, 0), (0, -1)}

d. A = {(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4}

4. Express the situation as a rule: "The area *A* of a square is a function of the length of its side *s*."

a. A = 4s b. $A = s^2$ c. $A = s^3$ d. $A = 4s^2$

5. If g(m) = 3m – 4 and h(m) = 6m, then what is (g o h) (m)?

a. 18m – 4 b. 18m – 14 c. 18m – 24 d. 18m – 34

6. Given that $f(x) = \sqrt{x}+3$ and $g(x) = \sqrt{x}-1$, find (f o g) (2).

a. 1 b. 2 c. 3 d. 4

7. A proposed fare for a public train would charge ₱20.00 for the first 8 km of travel and ₱3.00 for each additional kilometer over the proposed fare. Find the proposed fare for a distance of 50 km.

a. ₱116.00 b. ₱126.00 c. ₱136.00 d. ₱146.00

8. An open box is to be made from a rectangular piece of cardboard having dimensions 10 inches by 20 inches by cutting out identical squares of area x2 from each corner and turning up the sides. Express the volume V of the box as a function of x.

a. $V = x^3 - 60x^2 + 400x$

b. $V = 2x^3 - 60x^2 + 400x$

c. $V = 3x^3 - 60x^2 + 400x$

d. $V = 4x^3 - 60x^2 + 400x$

9. Which of the following equation represents a one-to-one function?

a. $y = 4$ b. $x = 3$ c. $y = 3x$ d. $y=3x2$

10. The equation $y = 3x – 5$ represents a one-to-one function. Which could be a valid reason for the statement?

a. Any straight lines represent one-to-one function.

b. Its graph will intersect any vertical lines at only one point.

c. Its graph will intersect any horizontal lines at only one point.

d. Its graph will intersect any vertical and horizontal lines at only one point.

MODULE1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master on how to solve limits of a functions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:

- Lesson 1.1 – Functions and its Representations
- Lesson 1.2 – Determining whether the relation represents a function or not
- Lesson 1.3 – Vertical Line Test

After going through this module, you are expected to:

1. represents real-life situations using functions, including piecewise functions.
2. determine whether the relation represents a function or not.
3. understand and apply vertical line test.

Lesson 1.1	Functions and its Representations

What is It

In mathematics and other disciplines, functions are occurring frequently in different forms. Some of the representations in real-life situations that make use of function are listed below:

a. The Perimeter P of a regular hexagon is a function of the length of its side s.

$P = 6s.$

b. The area *A* of a circle is a function of its radius r.

$A = \pi r^2$

c. The cost of text messaging where a user is charged ₱400 monthly for a particular mobile plan, which include 450 free text messages. Messages more than 450 are charged ₱1 each. Representing the monthly cost for text messaging using the function t(m), where m is the number of messages in a month, will give us:

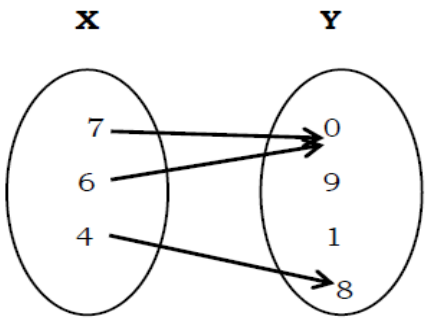
$$t(m)=\begin{cases} 400, 0<m\leq 450 \\ m-50, m>450 \end{cases}$$

In here, two pieces of functions must be considered to compute for the monthly cost. One is the t(m) = 400 for consuming 1 to 450 free text messages and the other is t(m) = 400 + m if in case reaches the limit and consumed more than 450 text messages.

Furthermore, **function** is a rule of correspondence between two nonempty sets, such that, to each element of the first set are called domain, there corresponds one and only one element of the second set called range.

The first elements in the ordered pairs (the *x*-values), form the domain. The second elements in the ordered pairs (the *y*-values), form the range. Only the elements "used" by the relation constitute the range.

To make it clearer, look closely to the diagram below.



The mapping diagram above shows a relation from Set X to Set Y,

where it consists of the ordered pairs (7,0) , (6,0) and (4,8).

The domain is the set {7,6,4}. The range is the set {0,8).

(Note that 9 and 1 are not part of the range, but part of the co-domain.)

Lesson 1.2	Determining whether the relation represents a function or not

Relations can be represented in five different ways:
•Ordered pairs
•Mapping Diagram
•Table of values
•Equation or rule
•Graph

Situation: Sean is working in a fast-food store as a crew. He earns P50.00 per hour.

From the situation,

If Sean worked for 6 hours, he earns P300.
If Sean worked for 7 hours, he earns P350.
If Sean worked for 8 hours, he earns P400.
And so on…

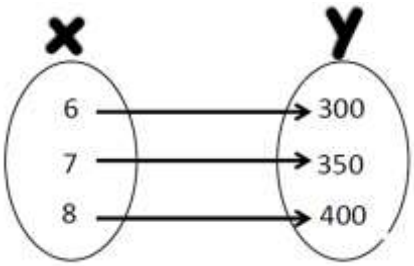
1. Representing the relation using ***Ordered Pairs*** we have,

(6, 300)
(7, 350)
(8, 400)

2.Representing the relation using ***Table of Values*** we have,

No. of Hours Worked (x)	6	7	8
Amount of Earning (y)	300	350	400

3. Representing the relation using ***Mapping Diagram*** we have,



Note: The elements in x represent the domain and the elements in y are the co-domain. Also, the relation depicts a one-to-one correspondence since the elements in X and Y is pairing of each object in X with one and only one element in Y. One-to-one correspondence illustrates a function since this relates only one value from the set of outputs (Y) to a value from set of inputs (X).

4. Representing the relation using ***Equation / Rule***, we have,

$$\boxed{y = 50x}$$

where x is the number of hours worked and y is the amount of his earning.

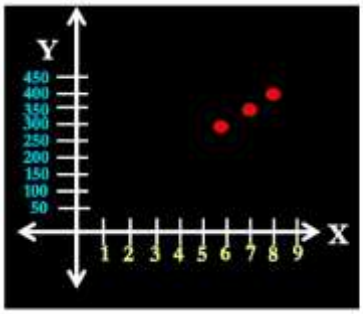
Also, in the equation $y = 50x$,

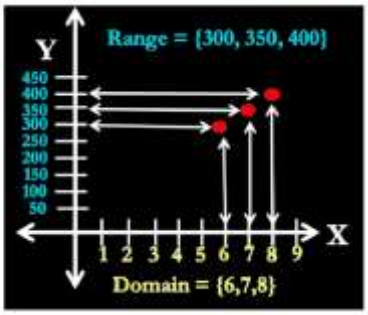
when x = 6 then y = 50(6) = 300.

when x = 7 then y = 50(7) = 350.

when x = 8 then y = 50(8) = 400.

5. Representing the relation using ***Graph***, we have,





Lesson 1.3 Vertical Line Test

What is It

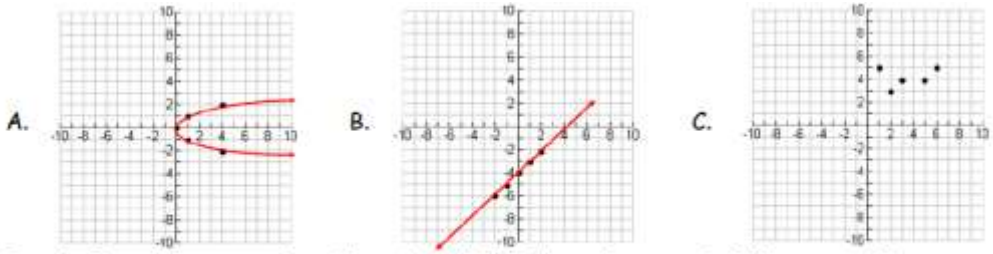
All functions are relations but not all relations depict functions.

To determine whether a relation represented in **ordered pairs and table of values** a function or not, just observe its x-values. If there is no repetition on the x-values, then it's a function otherwise, it is not.

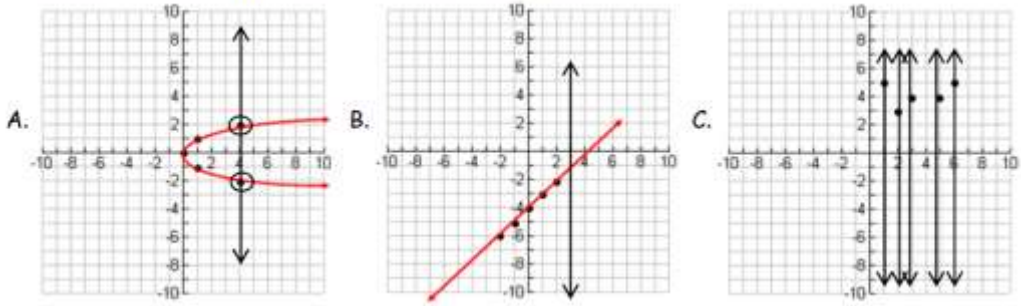
If the relation is expressed in **mapping diagram**, the one-to-one and many-to-one correspondence only illustrate a function. The one-to-many and many-to-many are not.

The relations when expressed in **equation/rule**, for every specific value of x, there should only be one corresponding y-value for that x-value. For example, y = x + 10, is a function because y will always be ten greater than x. Equations with exponents can also be functions. For example, $y = x^2-4$ is a function; although x-values of 2 and -2 give the same y-value of zero (0), that is only possible y-value for each of those x-values. However, $y^2 = x + 1$ is not a function because if you assign x-value of 8, y has two possible values which are 3 and -3. Expressing it in ordered pairs, (8, 3) and (8, -3). Looking at these ordered pairs, there is a repeated value in x.

Vertical Line Test is a way of testing to determine if a relation represented in **graph** is a function or not. A graph represents a function if and only if each vertical line intersects the graph at most once. Consider the following graphs below.



Graphs **B** and **C** are graphs of function while **A** is not because it did not pass the vertical line test or it intersects at two or more points.



What's More

Directions: Perform the following tasks below.

1. After transforming the ordered pairs in A into mapping diagram,
A = {(3, 6), (4,7), (3,8), (4, 1)}, which is true about it?
a. Set A illustrates a function.
b. Set A does not illustrate a function.
c. Set A illustrate a one-to-one correspondence
d. Set A illustrates a many-to-one correspondence

2. After plotting the set of ordered pairs in A = {(3, 6), (4,7), (3,8), (4, 1)}, which among the following is true about it?
a. The vertical line intersects at one point.
b. The vertical line intersects at two points.
c. The vertical line intersects at three points.
d. The vertical line intersects at four points.

3.Which among the image of parabolas that represent a function?
a. A parabola that opens to the left.
b. A parabola that opens to the right.
c. A parabola that opens upward.
d. A parabola that opens to left and right.

4.The set B = {(4,6), (8,6)} has two pairs with the same second coordinate. Which of the following is true about it?
a. Set B illustrates a function.
b. Set B does not illustrate a function.
c. Set B illustrate a one-to-one correspondence
d. Set A illustrates a many-to-one correspondence

5.Which of the following letters will pass the vertical line test when drawn on a Cartesian Coordinate Plane?
a.R b.S c.T d.U

What I Can Do

Read and study the situation below.

A rental home on EWD Apartelle rents for ₱1,000.00 a night for the first three nights, ₱900.00 for the next three nights, and ₱800.00 a night for each remaining night. The total cost T is a function of the number of night's n that a guest stays. Create a function model defined by function T.

MODULE 2

What I Need to Know

This module was designed and written with you in mind. It is here to help you master on Continuities and Discontinuities. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:

- Lesson 2.1 – Evaluating Functions
- Lesson 2.2 – Operations on Functions
- Lesson 2.3 – Solve Problems Involving Functions

After going through this module, you are expected to:

1. evaluate functions.
2. perform addition, subtraction, multiplication, division, and composition of functions.
3. solve problems involving functions.

Lesson 2.1 Evaluating Functions

What is It

Evaluating functions is replacing a given number or expression to its variable. To avoid mistake, just put parentheses ( ), on the replaced values.

Example 1: Evaluate the function $f(x) = 2x^2 + 3x + 1$ when $x = -4$.

Solution:

$f(x) = 2x^2 + 3x + 1$	Given function
$f(-4) = 2(-4)^2 + 3(-4) + 1$	Replace the variable "x" by "-4"
$= 2(16) - 12 + 1$	$(-4)^2 = 16$ and $3(-4) = -12$
$= 32 - 12 + 1$	$2(16) = 32$
$f(4) = 21$	$32 - 12 + 1 = 21$

Example 2: Evaluate the function $g(c) = 2c - c^2$ when $c = x + 1$.

Solution:

$g(c) = 2c - c^2$	Given function
$g(c) = 2(x + 1) - (x + 1)^2$	Replace the variable "c" by "x + 1"
$= 2x + 2 - (x^2 + 2x + 1)$	Using distributive property and FOIL Method
$= 2x + 2 - x^2 - 2x - 1$	$-(x^2 + 2x + 1) = -x^2 - 2x - 1$
$g(x + 1) = 1 - x^2$	Simplify

Example 3: If $h(x) = x^2 - x + 4$, find $h(m + n)$ and $h(m) + h(n)$.

Solution: Find $h(m + n)$

$h(x) = x^2 - x + 4$	Given function
$h(m + n) = (m + n)^2 - (m + n) + 4$	replace the variable c by "m + n"
$= (m^2 + 2mn + n^2) - (m + n) + 4$	Using FOIL Method and distributive property
$h(m+n) = m^2 + 2mn + n^2 - m - n + 4$	$-(m + n) = -m - n$ (distributive property)
$h(m+n) = m^2 + n^2 + 2mn - m - n + 4$	Arrange terms in descending order

Solution: Find h(m) + h(n)

$h(x) = x^2 - x + 4$	Given function
$h(m) = (m)^2 - (m) + 4$	replace the variable c by "m"
$h(m) = m^2 - m + 4$	Simplify
$h(n) = (n)^2 - (n) + 4$	replace the variable c by "n"
$h(n) = n^2 - n + 4$	Simplify
$h(m)+h(n) = (m^2-m+4)+(n^2-n+4)$	Add h(m) and h(n)
$h(m)+h(n) = m^2 + n^2 - m - n + 8$	Combine like terms

Evaluating Piecewise Functions

Functions in pieces are piecewise functions. There are piecewise functions which consist of two functions, three functions, four functions, and so on.

Beside these functions have their corresponding domains which give the restrictions and serve as clues to determine which function to be used for evaluation.

Example 1: In $y = \begin{cases} x - 2, & x \leq -2 \\ 3x^2 - 5, & x > -2 \end{cases}$

Find y when x = 7.

Solution: In here, there are two functions. The first is the y = x -2 and the second is the $y = 3x^2 - 5$. The domain of the first function is $x \leq -2$ which means that for all values of x that is less than or equal to -2, the function y = x-2 must be used. Then, use the second function $y = 3x^2 - 5$ when substituting values of x which are greater than -2.

To find y when x = 7, look at the domains. $X \leq -2$ does not include 7 but in $x > -2$. Therefore, use the second function that is opposite to $x > -2$ which is the $y = 3x^2 - 5$.

So,
$$y = 3x^2 - 5$$
$$y = 3(7)^2 - 5$$
$$y = 3(49) - 5$$
$$y = 147 - 5$$
$$y = 142$$

Example 2: In $y = \begin{cases} x - 2, & x \leq -2 \\ 3x^2 - 5, & x > -2 \end{cases}$ find y when x = -3.

Solution: In here, $X \leq -2$ includes -3. Therefore use the first function that is opposite to $x > -2$ which is the $y = x - 2$.

So,
$$y = x - 2$$
$$y = (-3) - 2$$
$$y = -3 - 2$$
$$y = -5$$

Example 3: In $y = \begin{cases} x - 2, & x \leq -6 \\ 3x^2 - 5, & 3 > x > -6 \\ x^2 - 2x + 1, & x \geq 3 \end{cases}$

Find y when x = 10.

Solution: In here, X ≥ 3 includes 10. Therefore use the function third function opposite to x ≥ 3, which is the $y = x^2 - 2x + 1$.

So, $y = x^2 - 2x + 1$

$y = (10)^2 - 2(10) + 1$

$y = 100 - 20 + 1$

$y = 81$

Lesson 2.2 Operations on Functions

What is It

Operations on functions

- involve addition, subtraction, multiplication, **division,** and composition of functions.

Functions, like numbers can be added, subtracted, **multiplied,** or divided because functions are usually given in equation **form,** and we perform these operations by applying them on algebraic expressions that appear on the right-hand side of the equations. Therefore, having the basic concepts on adding, subtracting, **multiplying,** and dividing algebraic expressions and factoring polynomials, is a great advantage in studying this topic.

Sum, Difference, Product, and Quotients of Functions

Let f and g be any two functions.

The sum f + g, difference f – g, product fg or f • g, and quotient $\frac{f}{g}$ are functions whose domain are the set of all real numbers common to the domain of f and f, are defined as follows:

1. Sum: $(f + g)(x) = f(x) + g(x)$
2. Difference: $(f - g)(x) = f(x) - g(x)$
3. Product: $(f \bullet g)(x) = f(x) \bullet g(x)$
4. Quotient $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

Example 1. If given the three functions below,

$f(x) = 7x - 4$	$g(x) = 2x + 3$	$h(x) = 4x^2 - 9$

Perform the following operations:

a. $(f + g)(x)$
b. $(g - f)(x)$
c. $(h + f - g)(x)$
d. $(f \bullet g)(x)$
e. $\left(\frac{h}{g}\right)(x)$

Solution: a. $(f + g)(x)$

$(f + g)(x) = f(x) + g(x)$	Sum of two functions
$= (7x - 4) + (2x + 3)$	Substitute the two functions
$= 9x - 1$	Combine similar terms

Solution: b. $(g - f)(x)$

$(g - f)(x) = g(x) - f(x)$	Difference of two functions
$= (2x + 3) - (7x - 4)$	Substitute the two functions
$= -5x + 7$	Combine similar terms

Solution: c. $(h + f - g)(x)$

$(h + f - g)(x) = h(x) + f(x) - g(x)$	Sum and Difference of functions
$=(4x^2 - 9) + (7x - 4) - (2x + 3)$	Substitute the three functions
$= 4x^2 + 5x - 16$	Combine similar terms

Solution: d. $(f \bullet g)(x)$

$(f \bullet g)(x) = f(x) \bullet g(x)$	Product of two functions
$=(7x - 4) \bullet (2x + 3)$	Substitute the two functions
$= 14x^2 + 13x - 12$	Product of two binomials

Solution: e. $\left(\frac{h}{g}\right)(x)$

$\left(\frac{h}{g}\right)(x) = \frac{h(x)}{g(x)}$	Quotient of two functions
$= \frac{4x^2 - 9}{2x + 3}$	Substitute the two functions
$= \frac{(2x-3)(2x+3)}{2x + 3}$	Factor using different of two squares
$= 2x - 3$	Simplify

Example 2. If $m(x) = 4x^2 + 3x - 5$ and $n(x) = 2x - 7$, compute for the value of:

a. $(m + n)(-3)$.
b. $(m \bullet n)(-3)$

Solution: a. $(m + n)(-3)$: Note that it is easier to evaluate first the given functions before performing the indicated operations

$(m + n)(-3) = m(-3) + n(-3)$	Sum of two functions
$m(-3) = 4(-3)^2 + 3(-3) - 5 = 22$	Evaluate the first function
$n(-3) = 2(-3) - 7 = -13$	Evaluate the second function
$m(-3) + n(-3) = 22 + (-13) = 9$	Add the values of the two functions

Solution: b. $(m \bullet n)(-3)$:

$(m \bullet n)(-3) = m(-3) \bullet n(-3)$	Product of two functions
$m(-3) = 4(-3)^2 + 3(-3) - 5 = 22$	Evaluate the first function
$n(-3) = 2(-3) - 7 = -13$	Evaluate the second function
$m(-3) \bullet n(-3) = (22)(-13) = -286$	Multiply the values of the two functions

Compositions of Functions

The composition of the function f with g is denoted by f o g and is defined by the equation

$$(f \circ g)(x) = f[g(x)]$$

The domain of the composition of function f o g is the set of all x such that

1. x is in the domain of g; and
2. g(x) is the domain of f.

The symbol f o g is read as "f circle g"

In computing for (f o g) (x), is replacing all x in f with g(x).

Example1. Given that $f(x) = 2x - 5$ and $g(x) = x^2 + 3$,

Find the following:

a. (f o g) (x)
b. (g o f) (x)
c. (f o f) (x)

Solution: a. (f o g) (x)

$f(x) = 2x - 5$	Given function
$(f \circ g)(x) = f[g(x)]$	Definition of Composition of Function
$= 2[g(x)] - 5$	Replace x by g(x)
$= 2(x^2 + 3) - 5$	Substitute g(x) by $(x^2 + 3)$
$= 2x^2 + 6 - 5$	Multiply $(x^2 + 3)$ by 2
$= 2x^2 + 1$	Simplify

Solution: b. (g o f) (x)

$g(x) = x^2 + 3$	Given function
$(g \circ f)(x) = g\,[f(x)]$	Definition of Composition of Function
$= [f(x)]^2 + 3$	Replace x by f(x)
$= (2x - 5)^2 + 3$	Substitute f(x) by (2x + 5)
$= 4x^2 - 20x + 25 + 3$	Multiply (2x - 5) by (2x – 5)
$= 4x^2 - 20x + 28$	Simplify

Solution: c. (f o f) (x)

$f(x) = 2x - 5$	Given function
$(f \circ f)(x) = f\,[f(x)]$	Definition of Composition of Functio
$= 2[f(x)] - 5$	Replace x by f(x)
$= 2(2x - 5) - 5$	Substitute f(x) by (2x - 5)
$= 4x - 10 - 5$	Multiply (2x – 5) by 2
$= 4x - 15$	Simplify

Lesson 2.3 Solve Problems Involving Functions

Solving Problems Involving Functions

A function is just like a machine that process inputs (domains) and produces outputs (range) depending on the rule that its operator instructs it to do.

For instance, in the function f(x) = 3x – 1, if you set an input for x = 7, then, when being processed by means of 2x – 1 and with correct computation, the output will become 20. So, when x = 7, then f(x) = 20.

These function machines are also related in real-life scenarios. Read and study the examples below:

Example:

A rental company charges a flat fee of ₱50.00 and an additional ₱10.00 per kilometer to rent a moving car. Write a function to approximate the cost y (in peso) in terms of x as the number of kilometers driven. How much would 75 km trip cost?

Solution: Consider the following,

For 10 km, y = 50 + 10(10) = 150
For 20 km, y = 50 + 10(20) = 250
For 30 km, y = 50 + 10(30) = 350

Therefore, the function model is represented by y=50 + 10x or y=10x + 50.
The function model can be used also to compute for 75 km trip cost,
For 75 km, y= 10(75) + 50 = 800. So, it costs ₱800.00 for travelling 75 km.

What's More

"The Group of Coronavirus"

Directions: Solve the following to decode and complete the statement about the group of coronaviruses.

Coronavirus is a big family of viruses and belong to

___	___	___	___	___	___	___	___	___		___	___	___	___	___	___.
2	**6**	**13**	**3**	**1**	**6**	**7**	**12**	**10**		**4**	**8**	**5**	**6**	**9**	**11**

Given that:

$f(x) = 4x - 5$ $g(x) = 3x + 2$ $h(x) = 5x^2 - 12$

1. Find f (5).	V. 15
2. What is g (-2)?	N. -4
3. Solve for f (0).	O. -12
4. Compute for (f + g) (3).	F. 56
5. Solve for (f o h) (1).	M. -33
6. Find (f + g – h) (-1).	I. -17
7. Compute for (g • h) (-3).	R. -231
8. Calculate for (f - h) (4).	A. -57
9. What is the value of g (20).	L. 62
10. Find g (40) – (f + h) (-2).	S. 127
11. Solve for g (9) • (f – g) (1).	Y. -174
12. Calculate for (h + h) (-4).	U. 136
13. What is (f – h + g) (3)	D. 29

14-15. Write the two words that you have formed.

What I Can Do

Read and study the situation below then answer the question that follows.

A car dealer offers a 20% discount on all news cars on his lot. At the same time, the auto manufacturer offers a ₱30,000 rebate.

a. Let P represent the sticker price of a car. Write a function f to represent its price if only the 30% discount is applied.

b. Write a g function to represent its price if only the ₱30,000 rebate is applied.

MODULE 3

What I Need to Know

This module was designed and written with you in mind. It is here to help you master on Derivatives The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:

- Lesson 3.1 – One-to-one Functions and its Inverse
- Lesson 3.2 – Graph of Inverse Functions
- Lesson 3.3 – Solve Problems Involving Inverse Functions

After going through this module, you are expected to:

1. represent real-life situations using one-to one function.
2. determine the inverse of a one-to-one function.
3. represent an inverse function through its:
   (a) table of values,
   (b) graph.
4. find the domain and range of an inverse function.
5. solve problems involving inverse functions.

Lesson 3.1 One-to-one Functions and its Inverse

What is It

The function is **one-to-one** when each y-value, there is just one x-value.

In symbol, $y = f(x)$

Note that, for each value of y in the range of f, there will only be one value of x in the domain of f. In other words, f is one-to-one if $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

Example 1: Determine whether the given relation is a function. If it is a function, determine whether it is one-to-one function.

a. The relation pairing of a student to his Learner's Reference Number (LRN).
b. The relation pairing of cellphone to its passcodes.
c. The relation pairing of the people to their personal experiences.
d. The relation pairing of a real number to its cube.
e. The relation pairing of online gamers to their favorite characters.

Solutions:

a. Each learner or student is assigned to one and only LRN. Therefore, the relation is a function. Also, two different students cannot be assigned the same LRN. So, the function is one-to-one.

b. Passcodes can be the same for cellphone A, cellphone B and to other cellphones. Representing it using mapping diagram, it depicts many-to-one correspondence. Therefore, the relation is a function. However, since two cellphones can be assigned with same passcodes, the relation pairing is not one-to-one.

c. In this world, everyone has its own individual personal experiences. Even twins have different personal experiences. Therefore, the relation pairing is a function and shows one-to-one.

d. Each real number has a unique perfect cube. Thus, the relation is a function. Also, each real number has unique value when it is being multiplied three times to itself or being cubed. For instance, 2 and -2, when 2 is being cubed, then its value is 8 and the cube of -2 is -8. So, since each real number (domain) will give a unique value when being cubed (range), then the relation pairing is one-to-one.

e. Mobile Legend is an example of an online game and some of the characters or heroes here are Cecilion, Pharsa, Nana, Hanzo, Estes and others. Let's consider the two gamers Gelo and Ian. In this scenario, it is possible that the favorite characters of Gelo are Cecilion and Nana. So, the relation is not a function. Also, Pharsa and Hanzo can be Ian's favorite. Since each gamer (domain) will not give a unique favorite character (range), then the relation pairing is not one-to-one.

Remember: A function has an inverse if and only if it is one-to-one.

Vertical Line Test: To test if the graph is a function; draw a vertical line on it. If it crosses at exactly one point, then the relation is a function.

Horizontal Line Test: This is being used to test if the graph shows one-to-one. If the horizontal line intersects at more than one point, then it is not one-to-one.

Horizontal and Vertical Line Test: If the horizontal line and vertical line intersect the graph at exactly one point, then it is one-to-one.

Example 2: Which between the two is a one-to-one?

a. $f(x) = 3x - 2$ b. $g(x) = x^2 + 1$

Solution: The function $f(x) = 3x - 2$ is one-to-one. If you assign x-value, there will always be one value for f(x). However, assigning values for x in the function $g(x) = x^2 + 1$ as 2 and -2, will give the same value for g(x) as 5. Therefore, the function $g(x) = x^2 + 1$ is not one-to-one.

The Inverse of One-to-one Functions

Properties of an inverse of a one-to-one function

Given a one-to-one function f(x) and its inverse $f^{-1}(x)$, then the following are true:

- The inverse of $f^{-1}(x)$ is f(x).
- $f(f^{-1}(x)) = x$ for all x in the domain of f^{-1}.
- $f^{-1}(f(x)) = x$ for all x in the domain of f.

For the second and third properties above, it can be imagined that evaluating a function f and its inverse f^{-1}in succession is like reversing the effect of the function.

Consider the functions $f(x) = 2x + 1$ and its inverse $f^{-1}(x) = \frac{x-1}{2}$.

In $f(x) = 2x + 1$ when x = **4,** then f(x) = 9. In $f^{-1}(x)= \frac{x-1}{2}$, when x = 9, then $f^{-1}(x)=$ **4**.

In $f(x) = 2x + 1$ when x = **5**, then f(x) = 11. In $f^{-1}(x)= \frac{x-1}{2}$, when x = 11, then $f^{-1}(x)=$ **5**.

In $f(x) = 2x + 1$ when x = **6**, then f(x) = 13. In $f^{-1}(x)= \frac{x-1}{2}$, when x = 13, then $f^{-1}(x)=$ **6**.

This means that $f\left(f^{-1}(x)\right) = x$ is true for all x in the domain of f^{-1}.

This also means that $f^{-1}(f(x)) = x$ is true for all x in the domain of f.

To find the inverse of $f(x) = 2x + 1$, which is $f^{-1}(x) = \frac{x-1}{2}$, study the solution below.

Example 1: Find the inverse of $f(x) = 2x + 1$.

$f(x) = 2x + 1$	Given function
$y = 2x + 1$	The equation of the function
$x = 2y + 1$	Interchange x and y variables

$y = \frac{x-1}{2}$	Solve for y in terms of x
$f^{-1}(x) = \frac{x-1}{2}$	Change y into $f^{-1}(x)$

Example 2: Find the inverse of f(x) = x + 1.

f(x) = x + 1	Given function
y = x + 1	The equation of the function
x = y + 1	Interchange x and y variables
y = x – 1	Solve for y in terms of x
$f^{-1}(x) = x - 1$	Change y into $f^{-1}(x)$

Example 3: Find the inverse of the rational function$g(x) = \frac{2x+3}{3x-5}$

Solution:

$g(x) = \frac{2x+3}{3x-5}$	Given function
$y = \frac{2x+3}{3x-5}$	The equation of the function
$x = \frac{2y+3}{3y-5}$	Interchange x and y variables
x (3y -5) = 2y +3 3xy – 5x = 2y + 3 3xy – 2y = 5x + 3 y (3x – 2) = 5x + 3 $y = \frac{5x+3}{3x-2}$	Solve for y in terms of x Multiply both sides by 3y -5 Multiply 3y – 5 by x Place all terms with y together Factor 3xy – 2y Divide both sides by 3x - 2
$g^{-1}(x) = x - 1$	Change y into $g^{-1}(x)$.

Lesson 3.2 Graphing Inverse Functions

What is It

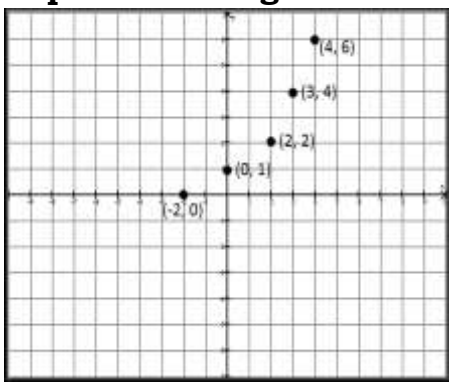
Graphing Inverse Functions

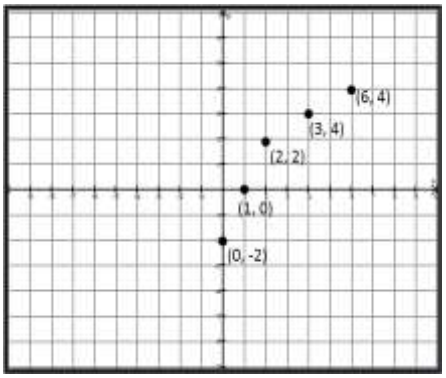
The graph of inverse function can be obtained by reflecting the graph about the line y = x. In other words, it is the reflection of the graph of the original function where the axis of symmetry is the line y = x.

Keep in mind that the graph must pass the horizontal and the vertical line test to declare that it shows one-to-one function and has an inverse function.

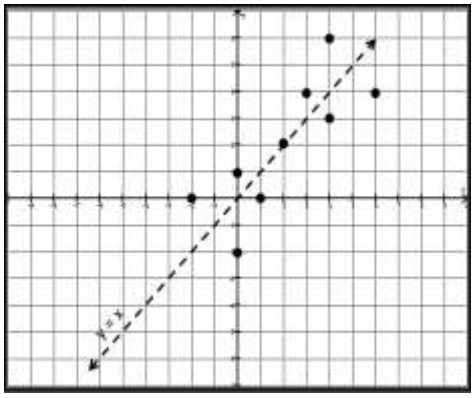
Example 1: The graphs of A = {(4, 6), (3, 4), (2, 2), (0, 1), (-2,0)} reflect its inverse A^{-1}= {(6, 4), (4, 3), (2, 2), (0, 1), (2, 0)} with respect to the line y = x.

Graph of the Original Function Graph of the Inverse Function





Graph of the Original and Inverse Functions



The graph of inverse function can be formed by graphing the original function and sketching the graph of its inverse.

Given the function f(x) $= -2x + 3$,

a. construct a table of values
b. generate the ordered pairs
c. sketch the graph of f(x)
d. find the domain and range of f(x)
e. Solve forf^{-1}(x)
f. construct a table of values for f^{-1}(x)
g. generate the ordered pairs for f^{-1}(x)
h. sketch the graph of f^{-1}(x)
i. find the domain and range of f^{-1}(x)

a. f(x) = -2x + 3

x	-3	-2	-1	0	1	2	3
f(x)	**9**	**7**	**5**	**3**	**1**	**-1**	**-3**

when x = -3
f(-3) = -2(-3)+3
= 6 + 3
= 9

when x = -2
f(-2) = -2(-2)+3
= 4 + 3
= 7

when x = -1
f(-1) = -2(-1)+3
= 2 + 3
= 5

when x = 0
f(-1) = -2(0)+3
= 0 + 3
= 3

when x = 1
f(1) = -2(1)+3
= -2 + 3
= 1

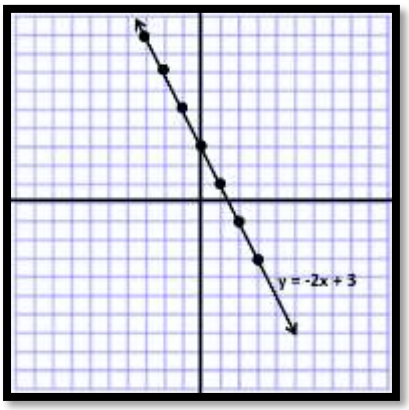
when x = 2
f(2) = -2(2)+3
= -4 + 3
= -1

when x = 3
f(3) = -2(3)+3
= -6 + 3
= -3

Note that the x-values can be continued until +∞ and -∞.
Therefore, it has also infinitely many y-values.

b. Using the table of values, ordered pairs can be generated as follows:
(-3 , 9) , (-2 , 7) , (-1 , 5) , (0 , 3) , (1 , 1) , (2 , -1) , (3 , -3)

c.



d. By looking at the graph, the domain of the function is the set of all real numbers. Also, the range is the set of real numbers since it is a linear function where its graph goes as high or as low without any limits. In interval notation,

Domain of f(x) : $D_f = (-\infty,+\infty)$

Range of f(x) : $R_f = (-\infty,-\infty)$

e. Solve for $f^{-1}(x)$

$f(x) = -2x + 3$	Given function
$y = -2x + 3$	The equation of the function
$x = -2y + 3$	Interchange x and y variables
$2y = -x + 3$ $y = \frac{-x+3}{2}$	Solve for y in terms of x
$f^{-1}(x) = \frac{-x+3}{2}$	Change y into $f^{-1}(x)$

f. $f^{-1}(x) = \frac{-x+3}{2}$

x	9	7	5	3	1	-1	-3
$f^{-1}(x)$	**-3**	**-2**	**-1**	**0**	**1**	**2**	**3**

when x = 9
$f^{-1}(9) = \frac{-9+3}{2}$
$= \frac{-6}{2}$
= -3

when x = 7
$f^{-1}(7) = \frac{-7+3}{2}$
$= \frac{-4}{2}$
= -2

when x = 5
$f^{-1}(9) = \frac{-5+3}{2}$
$= \frac{-2}{2}$
= 5

when x = 3
$f^{-1}(9) = \frac{-3+3}{2}$
$= \frac{0}{2}$
= 3

when x = 1
$f^{-1}(9) = \frac{-1+3}{2}$
$= \frac{2}{2}$
= 1

when x = -1
$f^{-1}(9) = \frac{-(-1)+3}{2}$
$= \frac{4}{2}$
= -1

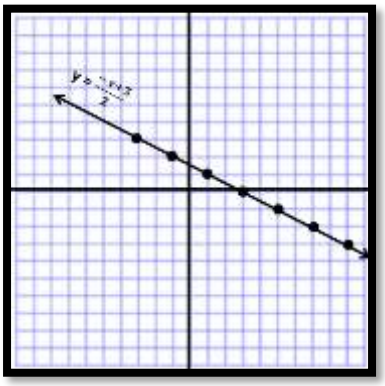
when x = -3
$f^{-1}(9) = \frac{-(-3)+3}{2}$
$= \frac{6}{2}$
= -3

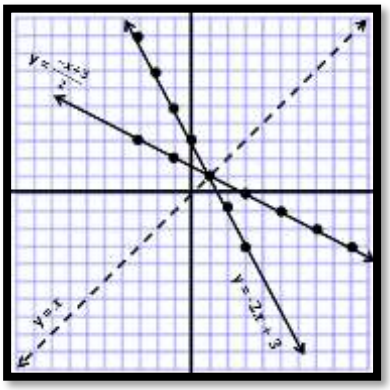
Note that the x-values can be continued until +∞ and -∞.

Therefore, it has also infinitely many y-values.

g. Using the table of values, ordered pairs can be generated as follows:
(9, -3), (7, -2), (5, -1), (3, 0), (1, 1), (-1, 2), (-3, 3)

h.





i. The domain of the function is the set of all real numbers. Also, the range is the set of real numbers. In interval notation,

Domain of $f^{-1}(x)$: $D_{f^{-1}} = (-\infty,+\infty)$

Range of $f^{-1}(x)$: $R_{f^{-1}} = (-\infty,-\infty)$

Also, the domain of the function is the range of its inverse, and the range of the function is the domain of its inverse.

- In graphing inverse function, follow the steps of graphing other functions by plotting points that belong to the inverse function and connect these points from left to right.
- Keep in mind that if (x, y) is a point from a given function then (y, x) is a point of the inverse of the function and the graph of the inverse of a function reflects the graph of the function with respect to the line y = x.
- The **domain** of a function f is the set of all values of x and the range of a function f is the set of all values of f(x) where x is an element of the domain of f.

Lesson 3.3 Solving Problems Involving Inverse Functions

What is It

Solving Problems Involving Inverse Functions

Applications of the inverse functions are problems involving reversible processes.

Example 1: You asked your friend to think of a number that is greater than zero. Add 4 to that number, square the sum, multiply the result by 9 and divide the result by 3. If the result is 363, find the number.

Solution: We first construct the function that will compute the final number based on the original number. Expressing the word problem into number sentence we come up with this function.

$$f(x) = (x + 4)^2 \bullet 9 \div 3 \text{ or } \qquad f(x) = \frac{9(x+4)^2}{3}$$

Looking at the function $f(x) = \frac{9(x+4)^2}{3}$, we can say that it is not a one-to-one function because it is in the form of quadratic functions where its graph is a parabola that opens upward. Parabola will not pass the horizontal and vertical line test. Also, assigning values for x in the function $f(x) = \frac{9(x+4)^2}{3}$ as 1 and -9, will give the same value for f(x) as 75. Therefore, the function is not one-to-one. However, since we restricted the domain as x is greater than zero (x > 0), we can say that it is one-to-one function, and we can find its inverse.

f. Solve for $f^{-1}(x)$

$f(x) = -2x + 3$	Given function
$y = -2x + 3$	The equation of the function
$x = -2y + 3$	Interchange x and y variables
$2y = -x + 3$ $y = \frac{-x+3}{2}$	Solve for y in terms of x
$f^{-1}(x) = \frac{-x+3}{2}$	Change y into $f^{-1}(x)$

What's More

"Inverse Mania"

The given function is $m(x) = x^2 - 3, x \geq 0$

1. Complete the Table of Values for $m(x) = x^2 - 3$

X	-2	-1	0	1	2
Y					

4. Solve for $m^{-1}(x)$ = _____

5. Complete the Table of Values for $m^{-1}(x)$

X					
Y	-2	-1	0	1	2

2. Give the Ordered Pairs from the table of values *in* m(x) = $x^2 - 3$

(,) (,) (,)

(,) (,)

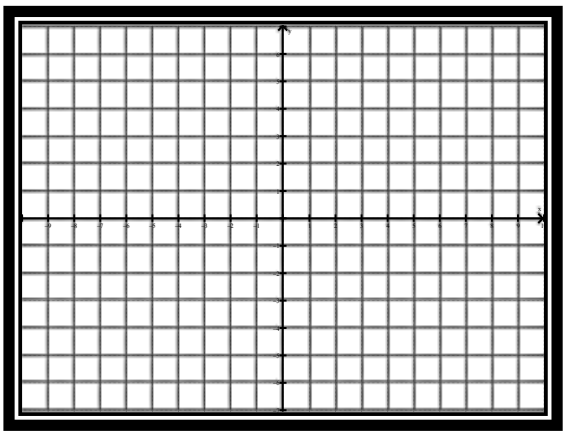
3. Plot the points on a Cartesian Plane.

6. Give the OrderedPairs from the table of values *in* $m^{-1}(x)$

(,) (,) (,)

(,) (,)

7.Plot the points on a Cartesian Plane then draw the line y=x.



8. Domain of m(x):

9. Range of m(x):

10. Domain of $m^{-1}(x)$:

11. Range of $m^{-1}(x)$:

12 – 15. Give a conclusion about the domain and range of the original function m(x) = $x^2 - 3$ to the domain and range of its inverse.

What I Can Do

Read and study the situation below then answer the question that follows.

Given the rational function f(x) = $\frac{x-4}{x+1}$

a. find the vertical asymptote of f(x).
b. find the horizontal asymptote of f(x).
c. find the inverse of f(x).
d. find the vertical asymptote of $f^{-1}(x)$.
e. find the horizontal asymptote of $f^{-1}(x)$.

MODULE 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you master on the differentiation rules. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:
Lesson 1 – Exponential Functions
Lesson 2 – Exponential Function, Equation, and Inequalities
Lesson 3 – Solving Exponential Equation and Inequalities

After going through this module, you are expected to:

1. Represent real-life situations using exponential functions.
2. Distinguish between exponential function, exponential equation, and exponential inequality.
3. Solve exponential equations and inequalities.

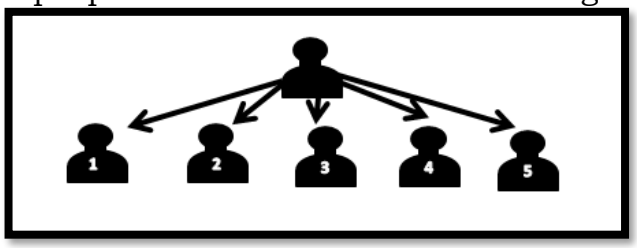
Lesson 4.1 Exponential Functions

What is It

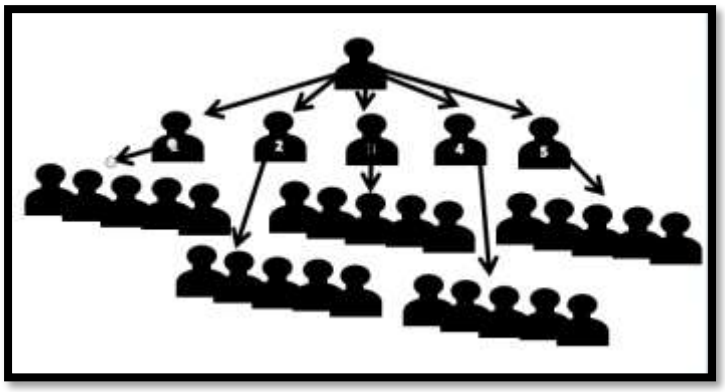
Situation A: A certain student is having an experiment about the number of chain message. In the conduct of it, he is going to send a message to 5 people through a messenger app in the first day of experiment. Then on the second day, these 5 people will send the same message to another 5 more new people and so on. How many people received the message on the tenth day?

To answer the problem above, consider the solution below.

On the first day, 5 people will receive the chain message.



Then each of them will send the same message to another 5 more new people as shown below. So, on the second day, 5(5) = 25 people.



If the pattern continues, these 25 people will send 5 messages to 5 more new people resulting to (25)(5)=125.

Question: What if the student wants to know how many people will receive the message on the:

a. 5th day b. 6th day c. 7th day d. nth day

Solution:

a. 5(5)(5)(5)(5)
   $= 5^5$
   = 3125
b. 5(5)(5)(5)(5)(5)
   $= 5^6$
   = 15,625
c. 5(5)(5)(5)(5)(5)(5)
   $=5^7$
   = 78,125
d. $y = 5^n$ or $f(x) = 5^n$ where y is the number of messages on n^{th} day.

Situation B: Peevee is a biology student from Rochean's University. He is trying to observe the behavior of the number of bacteria and decided to start his experiment with 5 bacteria in a petri dish. As he observes, he discovers that the number of bacteria triples after a day. To finish his experiment, he wants to find how many bacteria in that petri dish after 5 days.

Initially, there are 5 bacteria and after a day its number is getting tripled. Using a table of values, where y is the number of bacteria and x is the number of days, we can determine the number of bacteria in the petri dish after 5 days.

Days (x)	0	1	2	3	4	5
Bacteria (y)	5	15	45	135	405	1,215

We obtain 5 since it is the initial number of bacteria in a petri dish. After 1 day, 5 is getting tripled and becomes 15. Then on the second day, 10 is tripled resulting to 45. On the third day, the 45 bacteria tripled and become135. After 4 days, from 135 bacteria tripled again and its 405. On the last day, 405 bacteria times 3 is 1,215. Therefore, there will be 1,215 bacteria on his petri dish after 5 days.

Question: What if Peevee is going to extend his experiment and wants to know how many bacteria will be there after:

a. 6th day? b. 7th day? c. 8th day? d. nth day?

Solution:

a. 1,215 x 3
   = 3,645 bacteria (6th day)
b. 3,645 x 3
   =10,935 bacteria (7th day)
c. 10,935 x 3
   =32,805 bacteria (8th day)
d. $y = 5(3)^n$ or $f(x) = 5(3)^n$ where y is the number of bacteria on n^{th} day.

The situations above illustrate exponential functions since its function is in the form $f(x) = b^x$

Where:

- $b > 0$ and $b \neq 1$.
- x is a variable called exponent and
- b is a constant called base of the function.

Some of the examples of exponential functions are the following:

a. $f(x) = 4^x$
b. $y = 3(4)^x + 1$
c. $G(x) = A(1 + r)^t$
d. $y + 8 = 2^{x-4}$

Lesson 4.2 Exponential Function, Equation, and Inequalities

What is It

Exponential Function, Exponential Inequality and Exponential Equation

The table below will help you to distinguish exponential function, exponential inequality, and exponential equation. Take note that exponential expression is an expression that can be written in the form of $a(b)^{c+d}$ where $b > 0$ and $b \neq 1$.

Examples of exponential expressions are the following:

1. 4^x 2. $3(4)^x$ 3. $3(4)^{x+2}$ 4. $3(4)^{x+2} + 5$ 5. $3(4)^{x+2} - 5$

	Exponential Equation	Exponential Inequality	Exponential Function
Definition	An equation involving exponential expressions.	An inequality involving exponential expressions.	A function in the form of $f(x) = b^x$ where $b > 0$ and $b \neq 1$.
Examples	$4^{x+3} = 16$	$15(2)^x > 1500$	$f(x)=5(6)^{x+1} - 3$ or $y=5(6)^{x+1} - 3$
Examples	$12= 3(2)^x$	$3^x \leq 2187$	$g(x) = 5^x$ or $y = 5^x$

An exponential equation or inequality can be solved for all x values that satisfy the equation and inequality. An exponential function expresses a relationship between two variables (such as x and y) and can be represented by a table of values or a graph.

Example: Examine the given below then determine whether the given is an exponential function, exponential inequality, exponential equation, or none of these.

a. $y = 5(x)^2$
b. $3^x = 245$
c. $16 > 2^{x+1}$
d. $6^x = f(x)$
e. $y = (-3)^{x-2}$
f. $7 > 49^x$
g. $y = 5(2)^x$
h. $x^5 = 3125$
i. $32 \geq 4^x$
j. $y = (0.9)^x$

ANSWER	REASON
a. None of the these	The exponent is a fixed value.
b. Exponential Equation	It is an equation with exponential expression.
c. Exponential Inequality	It is an inequality with exponential expression.
d. Exponential Function	It is in the form of $f(x) = b^x$.
e. None of these	Since b is -3 and not greater than 0.
f. Exponential Inequality	It is an inequality with exponential expression.
g. Exponential Function	It is in the form of $y = b^x$.
h. None of these	The exponent is a fixed value.
i. Exponential Inequality	It is an inequality with exponential expression.
j. Exponential Function	It is an equation with exponential expression.

Lesson 4.3 Solving Exponential Equation and Inequalities

What is It

Solving Exponential Equations

An exponential equation in one variable is an equation having a variable in the exponent and can be solved for all values of that variable that will satisfy the equation. Take note of the following expressions:

$4^3= 64$	because 4 x 4 x 4 = 64	but	$4^{-3} = 1/64$
$3^4 = 81$	because 3 x 3 x 3 x 3 = 81	but	$3^{-4} = 1/81$
$2^3 = 8$	because 2 x 2 x 2 = 8	but	$2^{-3} = 1/8$

The one – to – one property of exponential functions will also help you in solving exponential equations.

One – to – one Property of Exponential Functions

If $x_1 \neq x_2$, then $b^{x_1} \neq b^{x_2}$, Conversely, if $b^{x_1} = b^{x_2}$, then $x_1 = x_2$

Example 1: What values of x that will make the equations true?
$2^x = 32$
$6^x = 216$
$1 = 23^x$
$14^x = 14$
$3^x = \frac{1}{81}$
Solution:
2 x 2 = 4
4 x 2 = 8
8 x 2 = 16
16 x 2 = 32
We multiplied 2 by itself 5 times. So, we can say that $2^5 = 32$. As you can see,

$2^5 = 32$
$2^x = 32$

to make the equation true, the value of x must be 5. Therefore, x = 5.
6 x 6 = 36
36 x 6 = 216
We multiplied 6by itself 3 times and we can say that $6^3 = 216$. So,

$6^3 = 216$
$6^x = 216$

to make the equation true, the value of x must be 3. Therefore, x = 3.
In here, $23^0 = 1$ since any number raised to zero except 0 is 1.
Therefore x = 0.
In $14^x = 14$, x must be 1 to make it true. Since $14^1 = 14$, then x = 1.
In $3^x = \frac{1}{81}$, having in your mind that $3^4 = 81$, but $3^{-4} = 1/81$. Therefore, x = -4.
The five examples above can be easily solved but for the next three examples, you may have to review the following concepts which are listed below.

Laws of Exponents

Laws	Notation	Examples
Zero Exponent	$x^0 = 1$	$5^0 = 1$, $(-5)^0 = 1$, $-5^0 = -1$
Negative Exponent	$x^{-1} = \frac{1}{x}$	$4^{-2} = \frac{1}{16}$, $-4^{-2} = -\frac{1}{16}$
Product of Powers	$(x^m)(x^n) = x^{m+n}$	$(2^3)(2) = 2^{3+1}$ $= 2^4$ $= 16$
Power of Product	$(xy)^m = x^m y^m$	$(3x)^2 = 3^2x^2$ $= 9x^2$
Power of a Power	$(x^m)^n = x^{m \cdot n}$	$(2^3)^4 = 2^{3 \cdot 4}$ $= 2^{12}$ $= 4{,}096$

Quotient of Powers	$\frac{x^m}{x^n} = x^{m-n}$	$\frac{4^5}{4^3} = 4^{5-3}$ $= 4^2$ $= 16$
Power of Quotient	$\left(\frac{x}{y}\right)^m = \frac{x^m}{y^m}$	$\left(\frac{4}{3}\right)^2 = \frac{4^2}{3^2}$ $= \frac{16}{9}$

Example 2: What values of x that will make the equations true?

a. $2^{2x-1} = 4^{3x+2}$
b. $4^{5x-13} = \frac{1}{8^x}$
c. $\frac{1}{125} = 25^{x+5}$

Solution:

a. To solve $2^{2x-1} = 4^{3x+2}$, our main target is to make their base to be equal.

$2^{2x-1} = 4^{3x+2}$	Given equation
$2^{2x-1} = (2^{2)3x+2}$	Transform 4 into 2^2
$2^{2x-1} = 2^{6x+4}$	Laws of Exponent: Power of a Power
2 = 2 2x – 1 = 6x + 4	One-to-one Property of Exponential Functions
2x-6x = 4+1 -4x = 5	Addition Property of Equality
$x = -\frac{5}{4}$	Divide both sides by -4

b. To solve $4^{5x-13} = \frac{1}{8^x}$ our main target is to make their base to be equal.

$4^{5x-13} = \frac{1}{8^x}$	Given equation
$(2^2)^{5x-13} = \frac{1}{(2^3)^x}$	Transform 4 into $2^2$and 8 into 2^3
$2^{10x-26} = \frac{1}{2^{3x}}$	Laws of Exponent: Power of a Power
$2^{10x-26} = 2^{-3x}$	Laws of Exponent: Negative Exponent
2 = 2 10x – 26 = -3x	One-to-one Property of Exponential Functions
10x + 3x = 26 13x = 26	Addition Property of Equality
x = 2	Divide both sides by 2

c.To solve $\frac{1}{125} = 25^{x+5}$our main target is to make their base to be equal.

$\frac{1}{125} = 25^{x+5}$	Given equation
$5^{-3} = (5^2)^{x+5}$	Transform 5 into -3and 25 into 5^2

$5^{-3}=5^{2x+10}$	Laws of Exponent: Power of a Power
5 = 5 -3 = 2x + 10	One-to-one Property of Exponential Functions
-2x = 10 + 3 -2x = 13	Addition Property of Equality
$x = -\frac{13}{2}$	Divide both sides by -2

Solving Exponential Inequalities

An exponential inequality in one variable is an inequality having a variable in the exponent and like exponential equation, it can be solved for all values of that variable that will satisfy the inequality.

Property of Exponential Inequalities

A. If b > 1, then the exponential function $y = b^x$ is increasing for all x. This means that $b^x<b^y$ if and only if x < y.

B. If 0 < b < 1, then the exponential function $y = b^x$ is decreasing for all x. This means that $b^x>b^y$ if and only if x < y.

Rules for Solving Inequalities

A. If the same real number is added to or subtracted from both sides of an inequality, the sense of the inequality is **not changed**.

B. If both sides of an inequality re multiplied by or divided by the same **positive** real number, the sense of the inequality is **not changed.**

C. If both sides of an inequality are multiplied by or divided by the same **negative** real number, the sense of the inequality is **CHANGED**.

Example 1: What values of x that will make the equations true?

a. $3^x < 9^{x-2}$

b. $\left(\frac{1}{3}\right)^{3x-2} > \left(\frac{1}{3}\right)^{3-2x}$

c. $\left(\frac{2}{5}\right)^{5x-1} \geq \left(\frac{4}{25}\right)$

a. To solve for x in $3^x < 9^{x-2}$, our main target is to make their base to be equal just like in exponential equation.

$3^x < 9^{x-2}$	Given inequality
$3^x < (3^2)^{x-2}$	Transform 9 into 3^2
$3^x < 3^{2x-4}$	Laws of Exponent: Power of a Power

3 = 3 x< 2x - 4	One-to-one Property of Exponential Functions
x - 2x <-4	Addition Property of Equality
-x <−4	Multiply both sides by -1
x > 4	Rules in Solving Inequality (C)

The values of x are all real numbers greater than 4 or $(4, \infty)$ in interval notation. Some of the values of x are 4.1, 4.99, 5, 5.5, 7, 100, 1000, and so on except 4 and less than 4.

b. In $\left(\frac{1}{3}\right)^{3x-2} > \left(\frac{1}{3}\right)^{3-2x}$, the bases are already equal.

$\left(\frac{1}{3}\right)^{3x-2} > \left(\frac{1}{3}\right)^{3-2x}$	Given inequality
$\frac{1}{3} = \frac{1}{3}$ 3x-2>3-2x	One-to-one Property of Exponential Functions
3x + 2x >3+2	Addition Property of Equality
5x>5	Divide both sides by 5
x <1	Rules in Solving Inequality (C)

Therefore, the values of x are all real numbers less than 1 or $(-\infty, 1)$ in interval notation. Some of the x values are 0.9, 0.8, 0, -1,-2, -3 and so on except 1 and greater than 1.

c. In $\left(\frac{2}{5}\right)^{5x-1} \geq \left(\frac{4}{25}\right)$, the bases must be equal by making $\frac{4}{25}$ into $\left(\frac{2}{5}\right)^2$.

$\left(\frac{2}{5}\right)^{5x-1} \geq \left(\frac{4}{25}\right)$	Given inequality
$\left(\frac{2}{5}\right)^{5x-1} \geq \left(\frac{2}{5}\right)^2$	Transform $\frac{4}{25}$ into $\left(\frac{2}{5}\right)^2$
$\frac{1}{3} = \frac{1}{3}$ 5x - 1≥ 2	One-to-one Property of Exponential Functions
5x ≥ 2+1	Addition Property of Equality
5x ≥ 3	Add 2 and 1
x ≥ $\frac{3}{5}$	Divide both sides by 5.
x ≤ 1	Rules in Solving Inequality (C)

Therefore, the values of x are all real numbers less than or equal to 1 or $(-\infty, 1]$ in interval notation. Some of the x values are 1, 2,3,4,5and so on except numbers which are greater than 1.

What's More

"The COVID-free Countries"

Directions: Solve the following to decode and reveal some countries which are COVID-free.

According to ALJAZEERA, here are few countries that have not reported any cases of coronavirus so far:

__	__	__	__	__	__	__	__	__	__	__	__	__	__	__	__
14	**6**	**3**	**6**	**15**	**1**	**7**	**6**	**12**	**1**	**9**	**10**	**1**	**7**	**10**	

__	__	__	__	__	__	__	__	__	__
2	**6**	**8**	**3**	**11**	**9**	**4**	**5**	**6**	**1**

__	__	__	__	__	__	__	__	__	__	__	__
7	**10**	**3**	**14**	**2**	**4**	**9**	**6**	**5**	**7**	**1**	**9**

__	__	__	__	__
7	**11**	**9**	**13**	**1**

1. $254^x = 1$
2. $9^x = \frac{1}{729}$
3. $4^x = 4096$
4. $7^x = 7$
5. $\frac{1}{5} = 5^x$
6. $2^x \geq 32$
7. $3^x < 2187$
8. $10^x \neq 100$
9. $81^x \geq 3^{2x+4}$
10. $5^{5x} < 5^{2x-9}$
11. $\left(\frac{1}{27}\right)^x = 3^{2-4x}$
12. $12^x > 1$
13. $81^{x+3} > 9^{3+x}$
14. $10^x = \frac{1}{100}$
15. $3^{2x} > 8$

K. $x = -2$
I. $x \geq 5$
R. $x = 4$
B. $x > 2$
A. $x = 0$
T. $x < 7$
M. $x = -3$
C. $x \neq 2$
O. $x = 2$
N. $x \geq 1$
E. $x = 1$
S. $x = -1$
U. $x < -3$
V. $x > 0$
G. $x > -3$

What I Can Do

"Solution Completion"

Directions: Complete the solution in solving the unknown variable for the following exponential equations.

1. $5^{x^2} = 5^{7x-12}$

$5 = 5$
$x^2 =$ _____
$x^2 - 7x + 12 = 0$
(____)(____) $= 0$
$x - 3 = 0$; _____ $= 0$
0
$x =$ ___ and $x =$ ___

2. $12^{3x^2+7x} = 144^{x+1}$

$12^{3x^2+7x} = (__)^{x+1}$
$12 = 12$
_____ = _____
$3x^2 + 7x - 2x - 2 = 0$
$3x^2 +$ ___ $- 2 = 0$

(_____)(_____) $= 0$
$3x - 1 = 0$; _____ $= 0$
$x =$ ____ and $x =$ _____

3. $169^{x^2-2x+9} = 13^{x^2+4x+2}$

$(__)^{x^2-2x+9} = 13^{x^2+4x+2}$
$13^{2x^2-4x+18} = 13^{x^2+4x+2}$
$13 = 13$
_____ = _____
$2x^2 - x^2 - 4x - 4x + 18 - 2 = 0$
_____ $= 0$
(_____)(_____) $= 0$
_____ $= 0$; _____ $= 0$
$x =$ ___ and $x =$ ____

MODULE 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you master on the Optimization (Minima-Maxima). The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into four lessons, namely:

- Lesson 5.1 –Representing an exponential function through its:

(a) table of values
(b) graph, and
(c) equation.

- Lesson 5.2– Determining the domain and range, intercepts, zeroes, and asymptotes of an exponential function.
- Lesson 5.3 - Solving problems involving exponential functions, equations, and inequalities.

After going through this module, you are expected to:

1. represent an exponential function through its:
   (a) table of values

(b) graph and
(c) equation

2. determine the domain and range, intercepts, zeroes, and asymptotes of an exponential function.
3. solve problems involving exponential functions, equations, and inequalities.

Lesson 5.1 Representing an exponential function through its: (a) table of values (b) graph, and (c) equation.

What is It

Representing Exponential Functions through its Table of Values, Graphs and Equation

An exponential function can be represented using table of values, graph, and equation. Look at the situation below on how to do this.

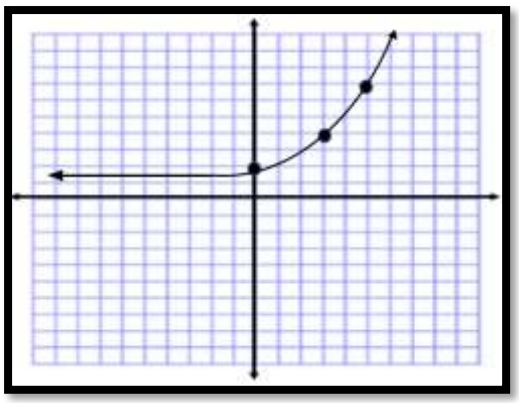
Situation: Supposed a scientist is observing the behavior of the number of 2 bacteria in a petri dish. If the number of bacteria doubles every 3 hours, how many bacteria are there after 15 hours?

From the situation above, there are 2 bacteria in the experiment. After 3 hours, it will be doubled so the number of bacteria will become 4. After 6 hours 4 bacteria will be doubled again and there will be 8 bacteria. There will be 16 bacteria after 9 hours, 32 bacteria after 12 hours. If the pattern continues, there will be 64 bacteria after 15 hours and so on.

The behavior of the number of bacteria can be reflected on the table of values as shown below.

No of Hours (x)	0	3	6	9	12	15
No. of Bacteria (y)	2	4	8	16	32	64

Since we already have a table of values, we can now plot these points on a Rectangular Coordinate Plane with points (0 , 2), (3 , 4) , (6 , 8) , (9 , 16) , (12 , 32) , (15 , 64).On its graph below, notice that the points (9 , 16) , (12 , 32) , (15 , 64) were not reflected. It is because of the limited space of the Rectangular Coordinate Plane.



To obtain the values from the table below using equation, use $y = 2(2)^{\frac{x}{3}}$ where the initial amount of 2 bacteria doubles every 3 hours.

x	0	3	6	9	12	15
y	2	4	8	16	32	64

when x = 0,

$y = 2(2)^{\frac{0}{3}}$

$y = 2(2)^0$

$y = 2(1)$

$y = 2$

when x = 3 (after 3 hours)

$y = 2(2)^{\frac{3}{3}}$

$y = 2(2)^1$

$y = 2(2)$

$y = 4$

when x = 6 (after 6 hours)

$y = 2(2)^{\frac{6}{3}}$

$y = 2(2)^2$

$y = 2(4)$

$y = 8$

when x = 9 (after 9 hours)

$y = 2(2)^{\frac{9}{3}}$

$y = 2(2)^3$

$y = 2(8)$

$y = 16$

when x = 12 (after 12 hours)

$y = 2(2)^{\frac{12}{3}}$

$y = 2(2)^4$

$y = 2(16)$

$y = 32$

when x = 15 (after 15 hours)

$y = 2(2)^{\frac{15}{3}}$

$y = 2(2)^5$

$y = 2(32)$

$y = 64$

Graphing Exponential Functions

Example 1: Consider the functions f(x) = 4^x and g(x) = $\left(\frac{1}{4}\right)^x$. Represent these exponential functions using table of values and graphs.

Solution: Assign values for x then compute for the values of f(x) and g(x). Since it is possible to replace any values for x, we can assign negative numbers too.

x	-3	-2	-1	0	1	2	3
f(x)	$\frac{1}{64}$	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16	64
g(x)	64	16	4	1	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{64}$

The solutions below are the process on how to obtain the following values for f(x) and g(x).

f(x) = 4^x

when x = -3,

$f(x) = 4^{-3}$

f(x) = 4^x

when x = -2

$f(x) = 4^{-2}$

f(x) = 4^x

when x = -1

$f(x) = 4^{-1}$

$f(x) = \frac{1}{4^3}$ $f(x) = \frac{1}{4^2}$ $f(x) = \frac{1}{4^1}$

$f(x) = \frac{1}{64}$ $f(x) = \frac{1}{16}$ $f(x) = \frac{1}{4}$

f(x) = 4^x f(x) = 4^x f(x) = 4^x f(x) = 4^x

when x = 0, when x = 1 when x = 2 when x = 3

$f(x) = 4^0$ $f(x) = 4^1$ $f(x) = 4^2$ $f(x) = 4^3$

$f(x) = 1$ $f(x) = 4$ $f(x) = 16$ $f(x) = 64$

g(x) = $\left(\frac{1}{4}\right)^x$ g(x) = $\left(\frac{1}{4}\right)^x$ g(x) = $\left(\frac{1}{4}\right)^x$

when x = -3, when x = -2 when x = -1

g(x) = $\left(\frac{1}{4}\right)^{-3}$ g(x) = $\left(\frac{1}{4}\right)^{-2}$ g(x) = $\left(\frac{1}{4}\right)^{-1}$

g(x) = $\left(\frac{4}{1}\right)^3$ g(x) = $\left(\frac{4}{1}\right)^2$ g(x) = $\left(\frac{4}{1}\right)^1$

g(x) = $(4)^3$ g(x) = $(4)^2$ g(x) = $(4)^1$

g(x) = 64 g(x) = 16 g(x) = 4

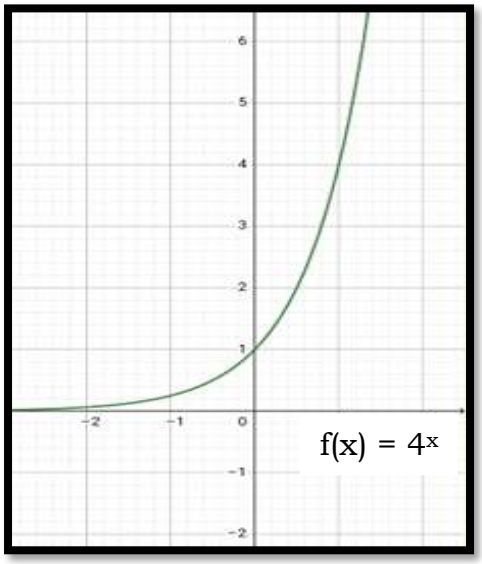
g(x) = $\left(\frac{1}{4}\right)^x$ g(x) = $\left(\frac{1}{4}\right)^x$ g(x) = $\left(\frac{1}{4}\right)^x$ g(x) = $\left(\frac{1}{4}\right)^x$

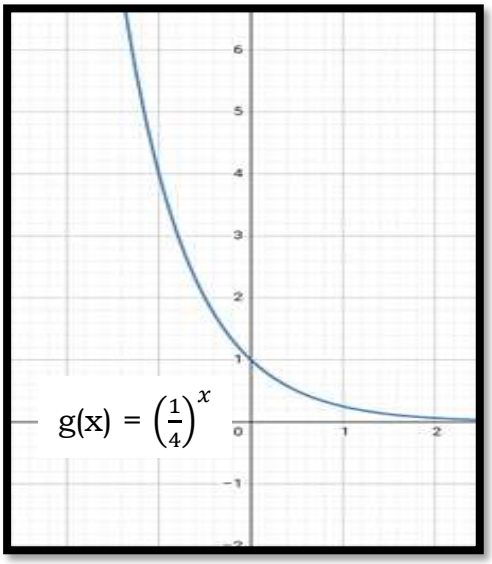
when x = 0, when x = 1 when x = 2 when x = 3

g(x) = $\left(\frac{1}{4}\right)^0$ g(x) = $\left(\frac{1}{4}\right)^1$ g(x) = $\left(\frac{1}{4}\right)^2$ g(x) = $\left(\frac{1}{4}\right)^3$

g(x) = 1 g(x) = $\frac{1}{4}$ g(x) = $\frac{1}{16}$ g(x) = $\frac{1}{64}$

The graph of f(x) = 4^xshows that it has an increasing trend while the graph of g(x) = $\left(\frac{1}{4}\right)^x$ has a decreasing trend. Also, we can see that f(x) = 4^xexhibits exponential growth since the base is greater than 1 while the other function g(x) = $\left(\frac{1}{4}\right)^x$ depicts exponential decay since the base is between 0 and 1.





Example 2: Find the exponential function represented by the table below:

x	-3	-2	-1	0	1	2	3
f(x)	$\frac{1}{2}$	1	2	4	8	16	32

Solution: The general form of an exponential function is $f(x) = ab^x$.

In the table of values above, when x = 0 , the value of f(x) is 4.

$$f(0) = ab^0 = 4$$

$$a(1) = 4$$

$$a = 4$$

When a is 1, we can write $f(x) = 4b^x$. If f(1) is 8, then

$$f(1) = 4b^1 = 8$$

$$4b = 8$$

$$b = \frac{8}{4}$$

$$b = 2$$

Therefore the exponential function represented by the table of values above is $f(x) = 4(2)^x$.

Lesson 5.2 Determining the domain and range, intercepts, zeroes, and asymptotes of an exponential function.

What is It

Intercepts, Zeroes, Asymptotes, Domain and Range of Exponential Functions

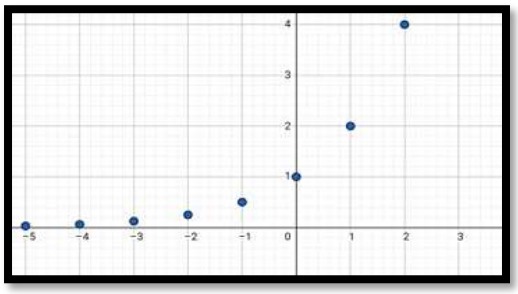
Example 1: Sketch the graph of $\boldsymbol{f(x) = 2^x}$.

Step 1: Construct a table of values to generate the ordered pairs of the points to be plotted on the Rectangular Coordinate Plane.

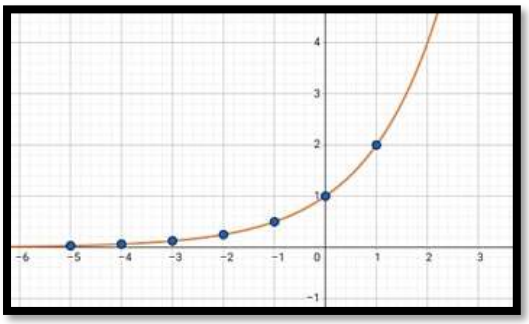
$$\boldsymbol{f(x) = 2^x}$$

x	-3	-2	-1	0	1	2	3
$f(x)$	1/8	1/4	1/2	1	2	4	8

Step 2: Plot the points that can be found in the table of values.



Step 3: Connect the points using a smooth curve.



As you sketch the graph of the function $\boldsymbol{f(x) = 2^x}$**,** notice that the function is defined for all values of x and increasing since it attains only positive y-values or the values of f(x). You can also observe that as the value of x decreases without bound, the function approaches to 0 but never touches it. So, the horizontal asymptote is y = 0.

The following are the characteristics of the function f(x) = 2^x:

Domain: Set of all real numbers or $(-\infty, +\infty)$ in interval notation.
Range: Set of all positive real numbers or $(0, +\infty)$ in interval notation.
y – intercept: 1 because the graph crosses the y – axis at point (0,1).
x – intercept: No x-intercept since the graph did not cross the x-axis.
Zero: There is no zero for the function since it did not cross the x-axis.
Horizontal Asymptote: y = 0
Vertical Asymptote: No vertical asymptote since all x-values is defined.

STEPS IN GRAPHING EXPONENTIAL FUNCTION

- Construct a table of values for the function.
- Plot the points on a Rectangular Coordinate Plane.
- Connect the points using a smooth curve.

PROPERTIES OF EXPONENTIAL FUNCTIONS

- Domain: Set of all possible values that the variable x can take. Look at the graph from left to right to see the x-values which are included.
- Range: Look at the graph from bottom to top to see the y-values which are included.

- **y – intercept: The intersection of the graph to the y-axis. To compute for this, set x = 0 then solve for y.**
- **x – intercept: The intersection of the graph to the x-axis. To compute for this, set y = 0 then solve for x.**
- **Zero: The x-value** when the function is equal to zero.
- **Horizontal Asymptote: a horizontal line where the graph** is approaching to.
- **Vertical Asymptote: a vertical line where the graph** is approaching to.

Lesson 5.3	Solving problems involving exponential functions, equations, and inequalities.

What is It

Problems Involving Exponential Functions, Equations, and Inequalities

Many real-life situations and word problems can be solved using exponential functions, exponential equations, and exponential inequalities.

Exponential Growth

Example 1:A certain town has a population of 50,000 that is increasing at the constant rate of 3% each year. Approximate the population of the town after 5 years.

Solution:

Initially there are 50,000 people and after a year, this will increase by 3% of 50,000 which will become 51,500. It happened because 3% of 50,000 or (0.03) (50,000) = 1,500. In short, (50,000) (1.3) = 51,500.

On the second year → (51,500) (1.03) = 53,045.
On the third year → (53,045) (1.03) = 54,636.35
On the fourth year → (54,636.35) (1.03) = 56,275.44
On the fifth year → (56, 275.44) (1.03) = 57, 963.70

Therefore, there will 57,963.70 or approximately 57,964 people after 5 years.

It can be computed using the formula $A = P(1+r)^t$ where:

A is the number of people
P is the initial or principal number of people
r is the annual rate
t is the number of years

Solving the problem using the formula above will give us the same answer.

Given that: A = ? P= 50, 000 r = 3% or 0.03 t = 5

$A = P(1+r)^t$
$A = 50{,}000(1 + 0.03)^5$
$A = 50{,}000(1.03)^5$
$A = 50{,}000\ (1.1592740743)$
$A = 57{,}963.70 \approx 57{,}964$ people

The formula above can be used also in solving problems that involve compounded interest.

Another application is the half-life of radioactive substance which refers to the time it takes for half of the substance to decay.

Half-Life of a Substance

Example 2: Suppose the half-life of a certain radioactive substance is 10 days and there are 10g initially. Determine the amount of substance remaining after 30 days.

Solution:

Initially, there are 10 g amount of substance. After 10 days, 10 g will become 5g since half of 10 is 5. Then After 20 days, 5g will become 2.5g and after 30 days, it will become 1.25g.

Or	at t = 0	Amount of Substance = 10g
	At t = 10days	Amount of Substance = 5g
	At t = 20days	Amount of Substance = 2.5g
	At t = 30days	Amount of Substance = 1.25g

By the use of the formula: $y = y0\left(\frac{1}{2}\right)^{\frac{t}{T}}$

where: y is the amount of substance after t units of time
y_0 is the initial amount of the substance at t = 0
T is the half-life of substance in T units.

In the problem, the given are listed below:

y_0= 10g
t = 30 days
T = 10 days
y = ?

Therefore, we have,

$$y = 10\left(\frac{1}{2}\right)^{\frac{30}{10}}$$
$$y = 10\left(\frac{1}{2}\right)^{3}$$
$$y = 10\left(\frac{1}{8}\right)$$
$$y = \frac{10}{8}$$
$$y = 1.25 \text{ g}$$

What's More

"The Most Common Symptoms"

Directions: Solve the following to decode and reveal the most common symptoms of Corona Virus Disease (COVID – 19)

Be careful of yourself and be informed the most common symptoms of COVID – 19.

__	__	__	__	__	__	__	__	__
67		**513**	**2**	**111**	**315**		**15**	

__ __ __ __ __ __ __ __

25 914128103

__ __ __ __ __

413113 5

Given that $f(x) = 3^x + 2$ $g(x) = 4^x - 3$ $h(x) = 2^{x-1}$

1. Domain of f(x), g(x) and h(x)	**U.** $(2, +\infty)$
2.y-intercept of g(x)	**N.** Increasing
3.Horizontal Asymptote of g(x)	**H.** $y = -3$
4. x-intercept of f(x)	**G.** $(-3, +\infty)$
5. Range of h(x)	**Y.** $y = 10$
6. y – intercept of f(x)	**E.** $(1, 1)$
7. The value of f(x) when x is 0.	**C.** 4
8. Range of f(x)	**V.** $(-\infty, +\infty)$
9. Horizontal Asymptote of h(x)	**S.** $\left(0, \frac{1}{2}\right)$
10. Range of g(x)	**O.** $y = 2$
11. Trend of f(x), g(x) and h(x)	**D.** $(0, -2)$
12. Horizontal Asymptote of f(x)	**F.** None
13. Intersection of g(x) and h(x)	**T.** $(0, 3)$
14. The value of h(x) when x is 4.	**I.** 3
15. y – intercept of h(x)	**R.** $(0, +\infty)$

What I Can Do

The characteristics of the function $f(x) = 3^{x-3} + 4$:

Domain:

Range:

y – intercept:

x – intercept:

Zero:

Horizontal Asymptote:

Vertical Asymptote:

Assessment

Multiple Choice. Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

1.It is a relation that relates each element of a set with exactly one element ofanother set.
A.ratio B.relation C.function D.abscissa

2.How will you determine whether the given relation in a form of table of valuesrepresents a function?
A.Compare the x-values. It is a function when it is being repeated.
B.Compare the y-values. It is a function when it is being repeated.
C.Compare the x-values. It is not a function when it is being repeated.
D.Compare the y-values. It is not a function when it is being repeated.

3.Which of the following set of ordered pairs does NOT represent a function?
A.{(a, b), (a, c), (a, d)} C.{(1, 2), (-1, 2), (0, 1)}
B.{(b, a), (c, a), (d, a)} D.{(4, 2), (2, 4), (1, 4)}

4.Which of the following correspondences of the elements in domain and rangerepresent a function?
A.one-to-one C.many-to-one
B.one-to-many D.A and C only

5.What is the value of m given that s(m) = 6m – 7 and s(m) = 11?
A.1 B.2 C.3 D.4

6. Find (g - f) (x) when f(x) = 3x – 4 and g(x) = 5x + 3.
A. 2x + 7 B. 2x – 7 C. 2x + 1 D. 2x – 1

7. In find y when x = 5.
A. -95 B. -29 C. 29 D. 95

8. Given the four functions below, create an expression that will produce a product of 455 when x = 8.
p(x) = 5-2x q(x) = 2x-3 r(x) = 6x s(x) = 4x+3
A. q(x)•p(x) B. q(x)•q(x) C. q(x)•r(x) D. q(x)•s(x)

9. To find the inverse of a function, you must interchange the variable x and y then solve for y. Given that $f(x) = 7x - 9$, perform the process above then solve for its inverse if it is one-to-one function.
A. $f^{-1}(x) = \frac{x+9}{7}$ C. $f^{-1}(x) = \frac{7x-9}{7}$
B. $f^{-1}(x) = \frac{x-9}{7}$ D. $f^{-1}(x) = \frac{-x-9}{7}$

10. Given that $f(x) = x^2 + 3$ where $x \geq 0$, examine the range below then select which one is correct about the range of its inverse.
A. $range = [0, +\infty)$ C. $range = (0, +\infty)$
B. $range = [3, +\infty)$ D. $range = (3, +\infty)$

Answer Key

Module 1

What I Know	Assessment
1. B 2. B 3. C 4. B 5. A 6. B 7. B 8. D 9. C 10.D	1. C 2. C 3. A 4. D 5. C 6. A 7. D 8. D 9. A 10.C

Module 2

What's More	What I Can Do
1.V 2.N 3.O 4.F 5.M 6.I 7.R 8.A 9.L 10.S 11.Y 12.U 13.D 14.NIDOVIRUS 15.FAMILY	It Varies

What's More	What I Can Do
1. B 2. B 3. C 4. A 5. D	It Varies

Module 5

What's More	What I Can Do
1. V 2. D 3. H 4. F 5. R 6. T 7. IU 8. U 9. Y 10. G 11. N 12. O 13. E 14. C 15. S	It Varies

Module 4

What's More	What I Can Do
1. A 2. M 3. R 4. E 5. S 6. I 7. T 8. C 9. N 10. U 11. O 12. V 13. G 14. K 15. B	It Varies

Module 3

What's More	What I Can Do
1. W 2. F 3. V 4. L 5. T 6. P 7. S 8. R 9. X 10. O 11. Y 12. E 13. I 14. N 15. M	It Varies

References

- Orines, F. B. (2016). New Century Mathematics 11 General Mathematics. Phoenix Publishing House. pp. 4-10
- Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc. pp. 1-7
- Orines, F. B. (2016). New Century Mathematics 11 General Mathematics. Phoenix Publishing House. pp. 38-54
- Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc. pp. 22-42
- General Mathematics Learner's Material. pp. 60-76
- Orines, F. B. (2016). New Century Mathematics 11 General Mathematics. Phoenix Publishing House. pp. 2-23.
- Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc.pp. 22-42
- General Mathematics Learner's Material. pp. 77-87
- Orines, F. B. (2016). New Century Mathematics 11 General Mathematics. Phoenix Publishing House. pp. 2-10.
- Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc.pp. 22-42
- General Mathematics Learner's Material. pp. 77-87
- Orines, F. B. (2016). New Century Mathematics 11 General Mathematics. Phoenix Publishing House. pp. 2-10.
- Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc.pp. 22-42

DIVISION OF NAVOTAS CITY

Senior High School

GENERAL MATHEMATICS

(Quarter 2)

First Semester

ADM
ALTERNATIVE DELIVERY MODE

S.Y. 2020-2021
NAVOTAS CITY PHILIPPINES

GOVERNMENT PROPERTY
NOT FOR SALE

General Mathematics for Senior High School
Alternative Delivery Mode
1st Semester (Quarter 2)
Second Edition, 2021

Republic Act 8293, section 176 states that: No copyright shall subsist in any work of the Government of the Philippines. However, prior approval of the government agency or office wherein the work is created shall be necessary for exploitation of such work for profit. Such agency or office may, among other things, impose as a condition the payment of royalties.

Borrowed materials (i.e., songs, stories, poems, pictures, photos, brand names, trademarks, etc.) included in this module are owned by their respective copyright holders. Every effort has been exerted to locate and seek permission to use these materials from their respective copyright owners. The publisher and authors do not represent nor claim ownership over them.

Published by the Department of Education
Secretary: Leonor Magtolis Briones
Undersecretary: Diosdado M. San Antonio

Development Team of the Module

Writer: Joseph G. Bantug, Angelo T. Leoncio, Marlon D. Silva, Patty Maryl C. Vergara

Editors: Maureen M. Neri, Jefferson S. Pabustan

Reviewers: Alberto J. Tiangco/Editha O. Peregrino

Illustrator: Joseph G. Bantug

Layout Artist: Joseph G. Bantug

Management Team: Alejandro G. Ibañez, OIC- Schools Division Superintendent
Isabelle S. Sibayan, OIC- Asst. Schools Division Superintendent
Loida O. Balasa, Chief, Curriculum Implementation Division
Alberto J. Tiangco, EPS in Mathematics
Grace R. Nieves, EPS In Charge of LRMS
Lorena J. Mutas, ADM Coordinator
Vergel Junior C. Eusebio, PDO II LRMS

Inilimbag sa Pilipinas ng _____

Department of Education – Navotas City
Office Address: BES Compound M. Naval St. Sipac-Almacen Navotas City

Telefax: 02-8332-77-64
E-mail Address: Navotas.city@deped.gov.ph

Table of Contents

What I Know 50

Module 1.....52

Module 2.....57

Module 3.....61

Module 4.....66

Module 5.....70

Module 6.....75

Module 7.....80

Module 8.....86

Assessment.....91

Answer Key 93

References 95

What I Know

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

1. What do you call the person or institution who invest the money or makes the fund available?
   A. Lender B. Borrower C. Principal D. B and C

2. What do you call the interest computed on the principal and on the accumulated past interest?
   A. Simple Interest C. Maturity Value
   B. Compound Interest D. Loan Date

3. Mark borrowed ₱75,500 at an annual simple interest rate of 7.5% for 3 years. How much interest should he pay?
   A. ₱16,987 B. ₱16,875 C. ₱19,250 D. ₱19,350

4. How much interest should he pay if he borrowed ₱20,800 at an annual interest rate of 1.8% for 2 years?
   A. ₱748.8 B. ₱758.8 C. ₱768.8 D. ₱738.8

5. It is a series of equal payments at regular intervals.
   A. Annuity C. Stocks
   B. Bond D. None of these choices

6. Find the present value of a man's pension of Php 13,000 payable monthly, the first due is at the end of 1 year, and the last at the end of 5 years, if money is worth 6% compounded monthly.
   A. Php 555,612.33 C. Php 825,612.33
   B. Php 645,612.33 D. Php 915,612.33

7. It is a type of investment that represents a share of ownership of a corporation.
   A. Bonds B. Market C. Lending D. Stocks

8. It is a money lent specially for business purposes.
   A. Business loan C. both A & B
   B. Consumer loan D. none of these choices

9. It is a declarative sentence that is either true or false, but not both.
   A. Proposition C. Compound Proposition
   B. Simple Proposition D. Conditional Proposition

10. Which among the following is true about the conditional of the proposition p and q?

A.

p	q	$p \rightarrow q$
T	T	F
T	F	F
F	T	T
F	F	F

C.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	F

B.

p	q	$p \rightarrow q$
T	T	T
T	F	T
F	T	F
F	F	T

D.

p	q	$p \rightarrow q$
T	T	F
T	F	T
F	T	F
F	F	T

MODULE1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master Illustrating Simple and Compound Interest. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module consists of lessons, namely:

- Lesson 1 – Illustrating Simple and Compound Interest

  Simple Interest

  Compound Interest

After going through this module, you are expected to:

4. Illustrate simple and compound interest.
5. Distinguishes between simple and compound interest.
6. Compute interest, maturity value, future value, and present value in simple interest and compound interest.

Lesson 1 Illustrating Simple and Compound Interest

What is It

Overview

Simple interest is the interest that is computed on the principal and then added to it, while Compound Interest is the interest computed on the principal and on the accumulated past interest.

Illustration of simple and compound Interest

Situation: Supposed you are planning to invest your ₱50,000.00 for 3 years. Bank A offers 2% simple interest per year, and Bank B offers 2% Compounded annually? Which will you choose and why?

Illustration of two investment

BANK A

Simple interest with an annual rate

Time (t)	Principal (P)	Simple Interest		Amount after t Years (Maturity Value)
		Solution	Answer	
1	₱50,000	(50,000) (0.02) (1)	1000	50,000+1000 = 51,000
2		(50,000) (0.02) (2)	2000	50,000+2000 = 52,000
3		(50,000) (0.02) (3)	3000	50,000+3000 = 53,000

BANK B

Compound Interest with an annual rate

Time (t)	Principal (P)	Simple Interest		Amount after t Years (Maturity Value)
		Solution	Answer	
1	50,000	(50,000) (0.02) (1)	1000	50,000+1000 = 51,000
2	51,000	(51,000) (0.02) (1)	1020	51,000+1020 = 52,020
3	52,020	(52,020) (0.02) (1)	1040.4	52,020+1040.4 = 53,060.4

Compare the Interests gained in the two investments

Simple Interest	Compound Interest
53,000 – 50,000 = 3,000	53,060.04 – 50,000 = 3,060.4

Let us Distinguish between the simple and compound interest base on the illustration.

SIMPLE INTEREST

The formula for finding Simple Interest:
I = Prt

Example 1

Given:
P = ₱100,000
r = 3% or 0.03
t = 5years
I =?

I = (100,000) (0.03) (5)

I = 15,000

Example 2

Given:
P =?
r = 2.5% or 0.025
t = 20years
I = 10,000

$$P = \frac{I}{rt}$$

$$P = \frac{10,000}{(0.025)(20)}$$

P = 20,000

Example 3
Given:
P = 50,000
r = 5% or 0.05
t = 10years
I = 25,000

$$r = \frac{I}{pt}$$

$$r = \frac{25{,}000}{(50{,}000)(10)}$$

r = 0.05 or 5%

Example 4
Given:
P = ₱10,000
r = 2% or 0.02
t = 30years
I = 6,000

$$t = \frac{I}{pr}$$

$$t = \frac{6000}{(10{,}000)(0.02)}$$

t = 30

Maturity (Future) Value

Situation: Maria wants to know the total amount of money in her bank account after 10 years at an interest rate of 5%. This amount is called Maturity Value or Future Value

Maturity (Future) Value
$F = P + I_c$
Where:
F = Maturity (Future) Value
P = Principal Amount
I_s= Simple Interest

Maturity (Future) Value
$F = P(1 + rt)$
Where:
F = Maturity (Future) Value
P = Principal Amount
r = Interest
t = term / time in years

Example 1:

Given:

P = ₱50,000

t = 10years

r = 5% or 0.05

F = P(1 + rt)

F = 50,000[1 + (0.05)(10)]

F = 50,000(1.5)

F = 75,000

Example 2:

Given:

P = ₱10,000

t = 20years

r = 3% or 0.03

F = P(1 + rt)

F = 10,000[1 + (0.03)(20)]

F = 10,000(1.6)

F = 16,000

Example 3:

Given:

P = ₱100,000

t = 5years

r = 2.5% or 0.025

F = P(1 + rt)

F = 100,000[1 + (0.025)(5)]

F = 100,000(1.125)

F = 112,500

COMPOUND INTEREST

Example:
Principal = ₱50,000
Int. rate = 5% compounded annually

Year	Amount at the end of the year	Amount at the end of the year
1	$P(1+r) = P(1+r)$	50,000·1.05 = 52,500
2	$P(1+r)(1+r) = P(1+r)^2$	52,500·1.05 = 55,125
3	$P(1+r)^2(1+r) = P(1+r)^3$	55,125·1.05 = 57,881.25
4	$P(1+r)^3(1+r) = P(1+r)^4$	57,125·1.05 = 60,775.31
5	$P(1+r)^4(1+r) = P(1+r)^5$	60,775.31·1.05 = 63,814.08

Formula for Maturity (Future) Value and Compound Interest

$$\boldsymbol{F = P(1+r)^t}$$

$F = Maturity\ (Future) value at\ the\ end\ of\ the\ term$
$P = Principal\ amount$
$t = term\ or\ time\ in\ years$
$r = interest\ rate$

Compound Interest

$$\boldsymbol{I_c = F - P}$$

Complete the table below by finding the unknown

Principal (P)	Rate (r)	Time (t)	Compound Interest (I)	Maturity Value (F)
20,000	3%	10 years	**(a)**	**(b)**
(c)	8%	5 years	**(d)**	100,000

Solution (b)

$$\boldsymbol{F = P(1+r)^t}$$
$$\boldsymbol{F = 20,000(1+0.03)^{10}}$$
$$\boldsymbol{F = 20,000(1.03)^{10}}$$
$$\boldsymbol{F = 26,878.33}$$

Solution (a)

$$\boldsymbol{I_c = F - P}$$
$$\boldsymbol{I_c = 26,878.33 - 20,000}$$
$$\boldsymbol{I_c = 6,878.33}$$

Solution (c)

$$\boldsymbol{P = \frac{F}{(1+r)^t}}$$
$$\boldsymbol{P = \frac{100,000}{(1+0.08)^5}}$$
$$\boldsymbol{P = \frac{100,000}{(1.08)^5}}$$
$$\boldsymbol{P = 68,058.32}$$

Solution (d)

$$\boldsymbol{I_c = F - P}$$
$$\boldsymbol{I_c = 100,000 - 68,058.32}$$
$$\boldsymbol{I_c = 31,941.68}$$

What's More

Activity: Complete the table below by finding the unknown

Principal (P)	Rate (r)	Time (t)	Interest (I)
10,000	3%	**(a)**	1500
20,000	5%	10	**(b)**
150,000	**(c)**	8	30,000
(d)	1.8%	10	18,000
250,000	3.5%	25	**(e)**

What I Can Do

Activity: Complete the table given the following choices

a. 67,195.82

b. 638,140.78

c. 2,093,777.93

d. 1,093,777.93

e. 138,140.78

f. 17,195.82

g. 366,095.71

h. 100,000

i. 56,269.56

j. 350,500

Principal (P)	**Rate (r)**	**Time (t)**	**Compound Interest (I)**	**Maturity Value (F)**
50,000	3%	10		
	8%	20		466,095.71
500,000	5%	5		
	1.5%	10		406,769.56
1,000,000	3%	25		

MODULE 2

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Simple and Compound Interest. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module covers only one lesson:

- Lesson 2 – Problem Solving Involving Simple and Compound Interest

After going through this module, you are expected to:

1. Solves problems involving simple and compound interest.

Lesson 2 Problem Solving Involving Simple and Compound Interest

What is It

Problem Solving Involving Simple Interest

Simple Interest When money is deposited in a bank or building society account, it commonly attracts interest; similarly, a borrower must normally pay interest on money borrowed. The rate of interest is usually (but not always) quoted as a rate percent per year. When writing a typical rate, it is 1.5% per annum for money deposited and 1%-2% per annum for money borrowed. Up-to-date rates are available from finance organizations.

There are two basic ways of calculating the amount of interest paid on money deposited: simple interest and compound interest.

If simple interest is paid, interest is calculated only on the principal *P*, the amount deposited (the original capital sum). The formula gives the interest *I* payable after one-year years at a rate of r% per annum.

Example:

Suppose Ericka borrowed ₱10,000 at an annual simple interest rate of 5% for 5 years. How much interest should he pay?

Principal Amount (P)	=	₱10,000
Interest rate (r)	=	5% or 0.05
Term or time in years (t)	=	5

Formula to find Simple Interest

$I = Prt$

$I = (10{,}000)\ (0.05)\ (5)$

$I = 2{,}500$

Find Interest(I), time(t), rate(r), and Principal amount (P) in simple interest

Problem 1

How much interest is charged when ₱20,000 is borrowed for 2 years at an annual interest rate of 3%?

Solution

Given:

P = ₱20,000
r = 3% or 0.03
t = 2
I =?

I =Prt
I = (20,000) (0.03) (2)

I = 1,200

Problem 2

Suppose Mark invested at an annual interest rate of 5%, which earned ₱37,500 of simple interest in 10 years. How much money was originally invested?

Solution

Given:

P =?
r = 5% or 0.05
t = 10years
I = 37,500

$$p = \frac{I}{rt}$$

$$p = \frac{37{,}500}{(0.05)(10)}$$

p = 80,000

Problem 3

How long will ₱30,000 amount to ₱45,000 if the simple interest rate is 10% per annum?

Solution

Given:

P = ₱30,000
r = 10% or 0.1
t =?
I = ₱15,000

$$t = \frac{I}{pt}$$

$$t = \frac{15{,}000}{(30{,}000)(0.1)}$$

t = 5

Problem 4

At what simple interest rate per annum will ₱150,500 accumulate to ₱80,000 in 20 years?

Solution

Given:

P = ₱150,500
r =?
t = 20 years
I = ₱60,200

$$r = \frac{I}{pt}$$

$$r = \frac{60{,}200}{(150{,}500)(20)}$$

r = 2%

Compound Interest

Simple interest is rarely used in real life: almost all banks and other financial institutions use compound interest.

This is when interest is added (or compounded) to the principal sum so that interest is paid on the whole amount. Under this method, if the interest for the first year is left in the account, the second year's interest is calculated on the whole amount so far accumulated.

Maturity (Future) Value

$F = P + I$

Where:

F = Maturity (Future) Value
P = Principal Amount
I = Simple Interest

Maturity (Future) Value

$F = P(1 + rt)$

Where:

F = Maturity (Future) Value
P = Principal Amount
r = Interest
t = term / time in years

Problem 1

What is the maturity value if ₱250,000 is invested at 3.5% compounded annually for 15 years?

Solution

$F = P(1 + rt)$

F = 250,000[1 + (0.035) (15)]

F = 250,000(1 + 0.525)

F = 250,000(1.0525)

F = 381,250

Problem 2

How much money do you need to place in a time deposit in a bank that pays a 5% interest rate compounded annually so that you will have ₱500,000 after 5 years?

Solution

$$P = \frac{F}{(1 + rt)}$$

$$P = \frac{500{,}000}{[1 + (0.05)(5)]}$$

$$P = \frac{500{,}000}{1 + 0.25}$$

$$P = \frac{500{,}000}{1.25}$$

What's More

Directions: Read and Solve the following problem.

1. ₱50,000 is borrowed for 5 years at an annual interest rate of 6%. How much is the interest?
2. Suppose Jun invested in a bank that gives a 5% interest rate per annum of 5%, an amount earned ₱150,000 in 5 years. How much money was originally invested?
3. How long will ₱50,000 amount to ₱100,000 if the simple interest rate is 5% per annum?
4. John invested ₱150,000 in a bank after 20 years. He earned ₱50,000. What is the interest rate?

What I Can Do

Directions: Read and Solve the following problem.

Mila invested ₱180,000 in a Bank that gives 3.7% compounded interest annually for 25 years. Find the following.

1. Total amount of his investment after 5 years
2. Total amount of his investment after 10 years
3. Total amount of his investment after 20 years
4. Total amount of his investment after 25 years

MODULE 3

What I Need to Know

This module was designed and written with you in mind. This is for you to understand and learn about Simple and General Annuities. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module consists of lessons, namely:

- Simple Annuity
- General Annuity

After going through this module, you are expected to:

1. Illustrate simple and general annuities (M11GM-IIc-1)
2. Distinguish between simple and general annuities (M11GM-IIc-2)
3. Find the future value and present value of both simple and general annuities (M11GM-IIc-d-1)

Lesson 3 Simple and General Annuities

What is It

Classification of Annuity based on Payment Schedule

1. Ordinary Annuity – is an annuity that is paid or received at the end of the time period.
2. Annuity Due – is an annuity that is paid or received at the beginning of the time period.
3. Deferred Annuity – is an annuity in which the first payment is delayed or deferred or deferred for a period of time.

Classification of Annuity based on Interest Period and Payment Interval

1. Simple Annuity – is an annuity in which the number of compounding periods per year coincides with the number of annuity payments per year.

2. General Annuity – is an annuity in which the annuity payments and compounding periods do not coincide. As was discussed in the preceding lesson, logarithms can be rewritten in exponential form. Recalling the exponential function $f(x) = b^x$ as taught in the previous lessons, we shall attempt to show that its inverse is a logarithmic function.

Lesson 1: SIMPLE ANNUITY

Simple annuity is an annuity where the payment interval is ***the same*** as the period of interest.

Future Value Formula

$$\boldsymbol{F_n = P_p\left[\frac{(1+i)^m - 1}{i}\right]}$$

F_n = Future value of an n-payment of ordinary simple annuity
P_p = Periodic payment / annuity
i = interest rate per period (r/K)
r = annual rate
K = no. of conversion periods in a year
t = no. of years
m = no. of conversion periods in a year x no. of years (K*t)

Present Value Formula

$$\boldsymbol{P_n = P_p\left[\frac{1-(1+i)^{-m}}{i}\right]}$$

P_n = Present value of an n-payment of ordinary simple annuity
P_p = Periodic payment / annuity
i = interest rate per period (r/K)
r = annual rate
K = no. of conversion periods in a year
t = no. of years
m = no. of conversion periods in a year x no. of years (K*t)

EXAMPLE 1. Suppose Mrs. Remoto would like to save P3,000 at the end of each month, for six months, in a fund that gives 9% compounded monthly. How much is the amount or future value of her savings after 6 months?

Solution

Given:
Pp = 3,000
i = 9% = $\frac{0.09}{12} = 0.0075$
m = 6 periods

Using the formula, we have.

$$F_n = P_p\left[\frac{(1+i)^m - 1}{i}\right]$$
$$= 3{,}000\left[\frac{(1+0.0075)^6 - 1}{0.0075}\right]$$
$$= 3{,}000\left[\frac{(1.0075)^6 - 1}{0.0075}\right]$$
$$= 3{,}000\left(\frac{1.0459 - 1}{0.0075}\right)$$
$$= 3{,}000\left(\frac{0.0459}{0.0075}\right)$$
$$= 3{,}000(6.12)$$
$$= \mathbf{18{,}360}$$

EXAMPLE 2. In order to save for her high school graduation, Marie decided to save P200 at the end of each month. If the bank pays 0.250% compounded monthly, how much will her money be at the end of 6 years?

Solution

Given:

Pp = 200

i = 0.250% = $\frac{0.0025}{12}$ = 0.0002

t = 6 years

m = tc = 6(12) = 72 periods

Using the formula, we have.

$$F_n = P_p\left[\frac{(1+i)^m-1}{i}\right]$$

$$= 200\left[\frac{(1+0.0002)^{72}-1}{0.0002}\right]$$

$$= 200\left[\frac{(1.0002)^{72}-1}{0.0002}\right]$$

$$= 200\left(\frac{1.0145-1}{0.0002}\right)$$

$$= 200\left(\frac{0.0145}{0.0002}\right)$$

$$= 200(72.5)$$

$$= \mathbf{14,500}$$

Lesson 2: GENERAL ANNUITY

General annuity is an annuity where the payment interval is ***not the same*** as the period of interest.

Future Value Formula

$$\boldsymbol{F_n = P_p\left[\frac{(1+i)^n-1}{f}\right]}$$

$$\boldsymbol{f = (1+i)^c - 1}$$

$$\boldsymbol{c = \frac{a}{b}}$$

F_n = Future value of an n-payment ordinary general annuity

P_p = Periodic payment / annuity

i = interest rate per period (r/K)

r = annual rate

K = no. of conversion periods in a year

t = no. of years

n = no. of conversion periods for the whole term (t*K)

f = effective interest rate per payment interval

c = number of conversation periods per payments interval

a = no. of months in a payment interval

b = no. of months in a compounding period

Present Value Formula

$$\boldsymbol{P_n = P_p\left[\frac{1-(1+i)^{-n}}{f}\right]}$$
$$\boldsymbol{f = (1+i)^c - 1}$$
$$\boldsymbol{c = \frac{a}{b}}$$

P_n = Present value of an n-payment ordinary general annuity
P_p = Periodic payment / annuity
i = interest rate per period (r/K)
r = annual rate
K = no. of conversion periods in a year
t = no. of years
n = no. of conversion periods for the whole term (t*K)
f = effective interest rate per payment interval
c = number of conversation periods per payments interval
a = no. of months in a payment interval
b = no. of months in a compounding period

EXAMPLE 1. Find the present value of an ordinary annuity of P2,000 payable annually for 9 years if the money is worth 5% compounded quarterly.

Solution

Given:
Pp = 2,000
t = 9
K = 4
n = 9(4) = 36
i = $\frac{r}{K} = \frac{0.05}{4} = 0.0125$
b = 3
a = 12
c = $\frac{a}{b} = \frac{12}{3} = 4$
f = $(1 + 0.0125)^4 - 1$
= 1.0509 – 1
f = 0.0509

Using the formula, we have.

$$P_n = P_p\left[\frac{1-(1+i)^{-n}}{f}\right]$$
$$= 2{,}000\left[\frac{1-(1+0.0125)^{-36}}{0.0509}\right]$$
$$= 2{,}000\left[\frac{1-(1.0125)^{-36}}{0.0509}\right]$$
$$= 2{,}000\left(\frac{1-0.6394}{0.0509}\right)$$
$$= 2{,}000\left(\frac{0.3606}{0.0509}\right)$$
$$= 2{,}000(7.0845)$$

EXAMPLE 2. P25,000 will be invested in an account at the end of each year at 4% compounded semi-annually. Find the size of the fund at the beginning of the 16th year.

Solution

Given:

Pp = 25,000

t = 15

K = 2

n = 15(2) = 30

$i = \frac{r}{K} = \frac{0.04}{2} = 0.02$

b = 6

a = 12

$c = \frac{a}{b} = \frac{12}{6} = 2$

$f = (1 + 0.02)^2 - 1$

$= 1.0404 - 1$

f = 0.0404

Using the formula, we have.

$$P_n = P_p\left[\frac{(1+i)^n - 1}{f}\right]$$

$$= 25{,}000\left[\frac{(1+0.02)^{30} - 1}{0.0404}\right]$$

$$= 25{,}000\left[\frac{(1.02)^{30} - 1}{0.0404}\right]$$

$$= 25{,}000\left(\frac{1.8114 - 1}{0.0404}\right)$$

$$= 25{,}000\left(\frac{0.8114}{0.0404}\right)$$

$$= 25{,}000(20.0842)$$

What's More

Practice Exercises:

A. Find the amount of each simple ordinary annuity.
   1. P2,000 deposited every year for 5 years at 9% per year compounded annually.
   2. P3,500 deposited every 6 months for 10 years at 8% per year compounded semi-annually.

B. Find the present value PV or the amount of future value FV of the general annuity.

	Payment	**Payment Interval**	**Term**	**Interest Rate**	**Find**
a.	P5,000	3 mo.	5 yrs.	5%, K = 12	FV
b.	P1,500	6 mo.	3 yrs.	8%, K = 2	FV
c.	P2,000	1 yr.	10 yrs.	6%, K = 4	PV
d.	P10,000	1 mo.	2 yrs.	10%, K = 4	PV
e.	P12,000	2 yrs.	15 yrs.	12%, K = 2	PV

What I Can Do

A. Use the formula $FV = P_p\left[\frac{(1+i)^n - 1}{i}\right]$ to calculate for the future values.

1. P = P4,000, i = 0.03, n = 4
2. P = P40,000, i = 0.05, n = 7
3. P = P2,500, i = 0.02, n = 6

B. The present value of an annuity of P5,000 every end of 3 months for 10 years when the interest rate is 4% compounded annually is P164,631.30. Use the given situation to give the values of the following variables:

1. P = _____
2. t = _____
3. K = _____
4. n = _____
5. r = _____
6. i = _____
7. b = _____
8. a = _____
9. c = _____
10. PV = _____

MODULE 4

What I Need to Know

This module was designed and written with you in mind. This is for you to understand and learn about Deferred Annuity. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two lessons, namely:

- Lesson 1 – Fair market value
- Lesson 2 – Deferred Annuity

After going through this module, you are expected to:

1. Calculate the fair market value of a cash flow stream that includes an annuity (M11GM-IId-2)
2. Calculate the present value and period of deferral of a deferred annuity (M11GM-IId-3)

Lesson 4 Deferred Annuity

What is It

Introduction

A deferred annuity in which the first payment is not made at the beginning nor at the end of the payment interval but a later date. The length of time when these payments are made is called the **period of deferment**. The first payment is made one period after the period of deferment. Thus, annuity that is deferred for 6 periods will have the first payment at the end of 7 periods. Likewise, in an annuity whose first payment is made at the end of the 7 periods, the annuity is deferred for 6 periods.

Let us discuss some examples of this type of annuity in real life.

• A credit card company offering its clients to purchase today but to start paying monthly with their choice of term after 3 months.
• A real estate agent is urging a condominium unit buyer to purchase now and start paying after 3 years when the condominium is ready for occupancy.
• A worker who has gained extra income now and wants to save his money so that he can withdraw his money monthly starting on the day of his retirement from work.

(a) Review definition of terms.

• **Annuity** - a sequence of payments made at equal (fixed) intervals or periods of time.
• **Annuity Immediate or Ordinary Annuity** - a type of annuity in which the payments are made at the end of each period.

(b) Present definition of terms.

• **Deferred Annuity** - an annuity that does not begin until a given time interval has passed.
• **Period of Deferral** - time between the purchase of an annuity and the start of the payments for the deferred annuity.

In contrast, the payments of a deferred annuity begin at the end of a later period.

Present Value of Deferred Annuity

$$\boldsymbol{P = A\left[\frac{(1+i)^{-d} - (1+i)^{-(n+d)}}{i}\right]}$$

Period of Deferral

$$\boldsymbol{d = \frac{logA\left[\frac{1-(1+i)^{-n}}{Pi}\right]}{log(1+i)}}$$

P_n = Present value of annuity
A = annuity
i = period of interest rate, i = r/K
n = number of payments in annuity, n = t(K)
d = number of compounding intervals in the period of deferral

EXAMPLE 1. On his 40th birthday, Mr. Ramos decided to buy a pension plan for himself. This plan will allow him to claim P10,000 quarterly for 5 years starting 3 months after his 60th birthday. What one-time payment should he make on his 40th birthday to pay off this pension plan, if the interest rate is 8% compounded quarterly?

Solution.

Given: $A = 10,000$ $r = 0.08$ $t = 5$ $K = 4$

Find: P The annuity is deferred for 20 years, and it will go on for 5 years. The first payment is due three months (one quarter) after his 60th birthday, or at the end of the 81st conversion period. Thus, there are 80 artificial payments.

Number of artificial payments: $d = Kt = 4(20) = 80$

Number of actual payments: $n = Kt = 4(5) = 20$

Interest rate per period: $i = \frac{r}{K} = \frac{0.08}{4} = 0.02$

Using the formula, we have.

$$\boldsymbol{P = A\left[\frac{(1+i)^{-d} - (1+i)^{-(n+d)}}{i}\right]}$$

$$= 10,000\left[\frac{(1+0.02)^{-80} - (1+0.02)^{-(20+80)}}{0.02}\right]$$

$$= 10,000\left[\frac{(1.02)^{-80} - (1.02)^{-(100)}}{0.02}\right]$$

$$= \mathbf{33,538.38}$$

Therefore, the present value of these monthly pensions is P33,538.38.

EXAMPLE 2. Find the present value of 10 semi-annual payments of P2,000 each if the first payment is due at the end of 3 years and money is worth 8% compounded semi-annually?

Solution.

Given:

A = P2,000

r = 8% = 0.08

K = 2

t = 5

i = $\frac{r}{K} = \frac{0.08}{2} = 0.04$

n = t(K) = 5(2) = 10

d = 5

Using the formula, we have.

$$\boldsymbol{P = A\left[\frac{(1+i)^{-d}-(1+i)^{-(n+d)}}{i}\right]}$$

$$= 2{,}000\left[\frac{(1+0.04)^{-5}-(1+0.04)^{-(10+5)}}{0.04}\right]$$

$$= 2{,}000\left[\frac{(1.04)^{-5}-(1.04)^{-(15)}}{0.04}\right]$$

What's More

Practice Exercises:

Find the period of deferral in each of the following deferred annuity problem (one way to find the period of deferral is to count the number of artificial payments)

1. Monthly payments of P2,000 for 5 years that will start 7 months from now
2. Annual payments of P8,000 for 12 years that will start 5 years from now
3. Quarterly payments of P 5,000 for 8 years that will start two years from now.
4. Semi-annual payments of P60,000 for 3 years that will start 5 years from now
5. Payments of P3,000 every 2 years for 10 years starting at the end of 6 years

What I Can Do

Answer the following problems completely.

1. Emma availed of a cash loan that gave her an option to pay P10,000 monthly for 1 year. The first payment is due after 6 months. How much is the present value of the loan if the interest rate is 12% converted monthly?
2. Adrian purchased a laptop through the credit cooperative of their company. The cooperative provides an option for a deferred payment. Adrian decided to pay after 4 months of purchase. His monthly payment is computed as P3,500 payable in 12 months. How much is the cash value

of the laptop if the interest rate is 8% convertible monthly?

3. Mr. and Mrs. Mercado decided to sell their house and to deposit the fund in a bank. After computing the interest, they found out that they may withdraw P350,000 yearly for 4 years starting at the end of 7 years when their child will be in college. How much is the fund deposited if the interest rate is 3% converted annually?
4. A group of employees decided to invest a portion of their bonus. After 3 months from today, they want to withdraw from this fund P5,000 monthly for 12 months to fund their gathering that they decide to do every month. How much is the total deposit now if the interest rate is 5% converted monthly?
5. Anna converted her loan to light payments which gives her an option to pay P1,500 every month for 2 years. The first payment is due 3 months from now. How much is the amount of the loan if the interest rate is 9% converted monthl

MODULE 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Stocks and Bonds. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. However, the order in which you read them can be changed to correspond with the textbook you are now using.

The module is consisting of the following competencies, namely:

- Illustrate stocks and bonds.
- Distinguishes between stocks and bonds.
- Describes the different markets for stocks and bonds.
- Analyzes the different market indices for stocks and bonds.

After going through this module, you are expected to:

1. Illustrate stocks and bonds.
2. Distinguishes between stocks and bonds.
3. Describes the different markets for stocks and bonds.
4. Analyzes the different market indices for stocks and bonds.

Lesson 5 Stocks and Bonds

STOCKS

Different companies offer different types of investment we can choose from. One of these types is STOCKS or also known as EQUITY. It represents the part ownership of a corporation. This entitles the stock owner to an equal ratio of the corporation's assets and profits equal to how much stock they own and how much capital gains and losses. Units of stock are called "shares." The investor may not get his original investment; dividends are optional and not an obligation.

Stocks may be purchased or sold at their current market value price. When a person owns such shares, the person gets a certificate containing the name of the company, the name of the purchaser, the number of shares, and the par value of the shares. Sources of return on securities are dividends and capital appreciation.

Participants in A Stocks Market

Investor is any person who puts or commits capital with the expectation of receiving financial returns, relies on different financial instruments to earn a significant return, and to manage financial objectives like building retirement savings, funding a college education, or merely accumulating additional wealth over time.

Stock Exchange is an institution regulating the stock market and facilitating the trading of shares by investors.

Brokers is an individual act as an agent or firm that acts as an intermediary between an investor and a securities exchange. It is also known as the fund manager,

Listed Companies A company whose shares are listed by the Stock Exchange as being available for buying and selling under the rules and safeguards of the Exchange. Figure 1. on the next page, show different listed companies in the Philippines.

Source: ("Blue chips: Philippines' best 30 companies in PSEi," 2020)

Figure 2. Top Blue Chips Companies in the Philippines

Types of Stocks:

1. **Preferred Stock** generally does not an opportunity in voting. The divided sharing is like fixed income, which is given priority if a company goes bankrupt and is liquidated.
2. **Common Stocks** are given the right to participate in voting, such as electing the company's board of directors, several votes equivalent to the number of shares. Enjoy the higher return of investment over years of market activities and receive any dividends paid out by the corporation.

Definition of Terms in Relations to Stocks

Stocks - Part-ownership

Dividend – Company's profit

Dividend per Share –the ratio of the dividends to the number of shares

Stock Market –a place where stocks are regulated, can be bought, and sold.

Market Value – unit price or current price of a stock at which it can be sold.

Par Value – the per-share amount, as stated on the company certificate. It is determined by the company and remains stable over time.

BONDS

It is an adept instrument, a certificate that proves that a company or corporation borrowed money from a certain group of individuals or investors for a definite period at a fixed rate. Bonds are used by companies, states, municipalities,

and governments to finance projects and operations. Owning a bond is just saving money from the bank in which you will earn interest from the amount being borrowed.

Participants in Bond

Investor is any person who puts or commits capital with the expectation of return or the lender or bondholder.

Debtholder or creditor is the borrower.

Bureau of the Treasury (BTr) acts as principal custodian of the national government's financial assets. It helps formulate policies on borrowing, investment, capital market development, managing cash resources, collecting taxes, and controlling and servicing public debt.

Definition of Terms in Relations to Bonds

Bond – Debt instrument 0r loan, amount of money on the maturity date.

Coupon – periodic interest payment that the bondholder receives during the time between purchase date and maturity date. Usually received semi-annually.

Coupon Rate – the rate per coupon payment period.

Price of a Bond – the price of the bond at purchase time.

Par Value or Face Value – the amount payable on the maturity date.

If P = F, the bond is purchased at par.
P < F, the bond is purchased at a discount.
If P > F, the bond is purchased at premium

Term of a Bond – fixed period (in Years) at which the bond is redeemable as stated in the bond certificate, number of years from the time of purchase to maturity date.

Fair Price of a Bond – the present value of all cash inflows to the bondholder.

Solve the following:

1. Trina is planning to have 600 shares in the stocks market, with a market price of P20.00 per share. How much will she pay to invest in the stock market?

   Given: no. of share = 600
   Market price = P20.00
   Find Par Value

SOLUTIONS:
Par Value = (No. of share) (Market price)
(600) (P20)
Par Value = P12000
Therefore, Par Value is P12000

2. A Certain financial institution declared a P 30,000,000 dividend for the common stocks. If there are 700,000 shares of common stocks, how much is the dividend per share?

   Given: Total Dividend = 30,000,000 Total Shares = 700,000
   Find: Dividend per share

   SOLUTIONS:
   Dividend per share = $\frac{\text{Total Dividend}}{\text{Total Shares}}$
   $\frac{30,000,000}{700,000}$
   Dividend per share = 42.28

   Therefore, the dividend per shares is 42.28

3. Find the yearly income of Mr. Miranda from an investment of 1000 shares of Blue Marquise stocks if a dividend of 15.60 per share is paid quarterly.

   Given: Par Value = 1000
   Divided per share = 15.60
   Find: Find the yearly income

   Find Total divided = (Par Value) (divided per share)
   (1000) (15.60) = 15600

   Yearly income = (Divided) (n0. Quarter in a year)
   (15600) (4) = 62400

   Therefore, the Yearly income is 62400

Directions: Read each statement or question below carefully and **fill in the blank**(s) with the correct answer.

1. _____ Company's profit
2. a place where stocks are regulated, can be bought, and sold _____,
3. It represents the part ownership of a corporation. Which entitles the owner of the stock to an equal ratio of the corporation's assets and profits equal to how much stock they own and how much capital gains and losses_____.

4. present value of all cash inflows to the bondholder. _____ .
5. _____ are used by companies, states, municipalities, and governments to finance project and operations. Owning a bond is just saving money from the bank in which you will earn interest from the amount being borrowed.

What I Can Do

Directions. Read and solve each problem.

1. A certain land developer declared a dividend of P 23,000,000 for its common stock. Suppose there are 500,000 shares of common stock. How much is the dividend per share?
2. A financial institution will give out a dividend of P57,000,000 for its common stock. Suppose there are 700,000 shares of common stock. How much is the dividend per share?
3. A resort company gave out a P11 dividend per share for its common stock. The market value of the stock is P17. Determine the stock yield ratio.
4. A telecommunication company gave out a P800 dividend per share for its common stock. The market value of the stock is 1180. Determine the stock yield ratio.

5. Find the amount of the semi-annual coupon for a P110,000 bond, which pays 4.5% convertible semi-annually.

MODULE 6

What I Need to Know

This module was designed and written with you in mind. This is for you to understand and learn about Business and Consumer Loans. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two lessons, namely:

- Lesson 1 – Business Loans and Consumer Loans
- Lesson 2 – Solving Problems on Business and Consumer Loans (Amortization and Mortgage)

After going through this module, you are expected to:

1. illustrate business and consumer loans. (M11GM-IIf-1)
2. distinguish between business and consumer loans (M11GM-IIf-2)
3. solve problems involving business and consumer loans (amortization, mortgage). (M11GM-IIf-3)

Lesson 6 Business and Consumer Loans

What is It

Introduction

Let us discuss the cost of a college education. Can you tell some success stories of how some college students were able to graduate because of the parents' sacrifices, or the students' dedication by resorting to having part-time jobs? Do you know that some families choose to borrow money from the banks or other lending institutions? Do you know how to borrow money from banks or other lending institutions? This will be the topic of our discussion for this module.

Definition of terms

• **Amortization Method** - method of paying a loan (principal and interest) on installment basis, usually of equal amounts at regular intervals
• **Mortgage** - a loan, secured by a collateral, that the borrower is obliged to pay at specified terms.
• **Chattel Mortgage** - a mortgage on a movable property
• **Collateral** - assets used to secure the loan. It may be a real-estate or other investments
• **Outstanding Balance** - any remaining debt at a specified time

Consumer loans are loans given to individuals for personal or family purpose while business loans are loans for businesses. Consumer loans and business loans may require a collateral. The collateral for both loans may be real estate or other investments. For business loans, they can use equipment, fixtures, or furniture as collateral. Consumer loans do not usually require a guarantor.

Business loans require the business owners to sign as guarantors. For consumer loans, the bank or the lending institution may require a credit report, bank statements, and an income tax return, and if the lendee is employed, a certificate of employment and employee pay slips. For business loans, the lendee must submit a

credit report, income tax returns and company's financial statement. The term of the business loan is generally shorter than the consumer loan. The interest rate for the business loan is usually higher than that for the consumer loan.

In Examples 1-5, identify whether the following is a consumer loan or business loan.

EXAMPLE 1. Mr. Agustin plans to have a barbershop. He wants to borrow some money from the bank for him to buy the equipment and furniture for the barbershop.

Answer: Business loan

EXAMPLE 2. Mr. and Mrs. Craig want to borrow money from the bank to finance the college education of their son.

Answer: Consumer loan

EXAMPLE 3. Mr. Alonzo wants to have some improvements on their 10-year-old house. He wants to build a new room for their 13-year-old daughter. He will borrow some money from the bank to finance this plan.

Answer: Consumer loan

EXAMPLE 4. Mr. Samson owns a siomai food cart business. He wants to put another food cart on a new mall in the other city. He decided to have a loan to establish the new business.

Answer: Business loan

A **mortgage** is a business loan or a consumer loan that is secured with a collateral. **Collaterals** are assets that can secure a loan. If a borrower cannot pay the loan, the lender has a right to the collateral. The most common collaterals are real estate property. For business loans, equipment, furniture, and vehicles may also be used as collaterals. Usually, the loan is secured by the property bought. For example, if a house and lot is purchased, the purchased house and lot will be used as a mortgaged property or a collateral. During the term of the loan, the **mortgagor**, the borrower in a mortgage, still has the right to possess and use the mortgaged property. If the mortgagor does not make regular payments on the mortgage, the **mortgagee** or the lender in a mortgage can repossess the mortgaged property. The most common type of mortgage is the **fixed-rate mortgage** wherein the interest remains constant throughout the term of the loan.

One of the most important and most common applications of annuities in business is the repayment of interest-bearing debts.

Amortization is a financial arrangement whereby a lump sum is incurred at compounded interest now, such as loan, and is liquidated or paid off or by a series of equal periodic payment for a specified amount of time.

Amortization of loan is the repayment of a loan by periodic payments, with possible exception of the last payment, are equal in amount. These payments are often described as combined payments of principal and interest. When the debt is

paid in series of periodic payments, pay the interest outstanding at the time payments are made and repay a part of the principal.

Amortization period is the length of time over which a loan is scheduled to be fully repaid, after the two parties (lender and borrower) agrees on the amount of a loan, the rate of interest, and the repayment frequency. After of which the amortization schedule will be established.

Amortization Schedule is a list os several periods of payments showing the principal and the interest parts of those payments and the outstanding balance after each payment is made.

<u>Formula in solving amortization</u>

$$\boldsymbol{P_n = A\left[\frac{1-(1+i)^{-n}}{i}\right]}$$

Pn = Present value of an n-payment ordinary simple annuity

A = Annuity / Periodic payment

r = rate in percent

K = compounding period

t = time in years

i = period of interest rate, i = r/K

n = number of payments in annuity, n = t(K)

EXAMPLE 1. Mr. Garcia borrowed P1,000,000 for the expansion of his business. The effective rate of interest is 7%. The loan is to be repaid in full after one year. How much is to be paid after one year?

Solution:

Given: P = 1, 000, 000 r = 0.07 t = 1

Find: F

$$\boldsymbol{F = P(1+r)^t}$$

$$= 1,000,000(1 + 0.07)$$

$$= 1,070,000$$

An amount of P1,070,000 must be paid after one year.

EXAMPLE 2. (Chattel mortgage). A person borrowed P1,200,000 for the purchase of a car. If his monthly payment is P31,000 on a 5-year mortgage, and the total amount of interest.

Solution:

Given: P = 1, 200, 000 Monthly payment = 31, 000

Find: Total Interest The total amount paid is given by

Total Amount = (31, 000) (12 months) (5 years) = 1, 860, 000
Thus, the total interest is the difference between the total amount paid and the amount of the mortgage:
Total Interest *= 1, 860, 000 − 1, 200, 000 =* ***660, 000***

What's More

Practice Exercises:

A. Identify the following whether it is a business loan or a consumer loan.
   1. Mr. Lim wants to have another branch for his cellphone repair shop. He decided to apply for a loan that he can use to pay for the rentals of the new branch.
   2. Trillas runs a trucking business. He wants to buy three more trucks for expansion of his business. He applied for a loan in a bank.
   3. Mrs. Alonzo decided to take her family for a vacation. To cover the expenses, she decided to apply for a loan.
   4. Glenn decided to purchase a condominium unit near his workplace. He got a loan worth P2,000,000.
   5. Mr. Galang purchased renovated her house P80,000. This was made possible because of an approved loan worth P75,000.

What I Can Do

Answer the following problems:

1. A loan of P200,000 is to be repaid in full after 3 years. If the interest rate is 8% per annum. How much should be paid after 3 years?
2. For a purchase of a house and lot worth P3,800,000, the bank requires 20% down payment, find the mortgaged amount.
3. A car dealer offers a 15% down payment for the purchase a car. How much is the mortgaged amount if the cash value of the car is P1,500,000?
4. Ms. Saya got a business loan worth P900,000. She promised to pay the loan semi-annually in 3 years. The semi-annual payment is P183,026.37 if money is worth 12% converted semi-annually. How much is the outstanding balance after the first payment?

5. A consumer loan worth P30,000 is to be repaid in 12 months at 9% convertible monthly. How much is the monthly payment?

MODULE 7

What I Need to Know

This module was designed and written with you in mind. It is here to help you master illustrating proposition and its kind. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:

- Lesson 1 – Illustrating and symbolizing propositions
- Lesson 2 – Distinguishing simple and compound propositions.
- Lesson 3 – Performing the different types of operations on propositions.

After going through this module, you are expected to:

1. Illustrates and symbolizes propositions
2. Distinguishes between simple and compound propositions.
3. Performs the different types of operations on propositions.

Lesson 7 Illustrating Propositions

What is It

PROPOSITION

Proposition is a declarative sentence that can be classified as true or false, but not both. But what is a declarative sentence?
Declarative sentence is used to make statement of fact, wish, intent, or feeling.

Example: I wish I could visit my grandparents this winter.

Be careful to differentiate the declarative sentence, imperative sentence, interrogative sentence, and exclamatory sentence. Carefully read its difference with the declarative sentence.

Imperative sentence is used to state a command, request, or directions. Example: Please read your lesson.

Interrogative sentence is used to ask a question and uses this mark (?) or called as the question mark. "What is your name?" is an example of this kind of sentence.

Exclamatory sentence is used to express strong feelings. "Get out of here!" is an example of this. Also, this kind of sentence uses exclamation point.

Note that if the given statement is a declarative sentence whether it is true or false is considered as proposition.

Examples: Determine whether each of the following is a proposition or not.

1. The number 5 is an odd number and greater than 3.
2. Marikina is the capital of the Philippines.
3. Do you have already your USB – OTG?
4. Your answer is correct.
5. That's great!

Solution:

1. The statement is true and a proposition.
2. The statement is false and a proposition too.
3. This is not a proposition since it is a question.
4. This is not a proposition since it is a paradox. Though it is a declarative sentence, we cannot consider that it is a proposition because we cannot tell if it is true or false.
5. This is not a proposition since it is a command or an imperative sentence.

Simple and Compound Propositions

Simple proposition is a proposition that conveys one thought with no connecting words whereas compound proposition contains two or more simple propositions that are put together using connective words.

Compound proposition uses some logical connectors involving proposition p and/or q may be expressed as follows:

not p	(negation)
p and q	(conjunction)
p or q	(disjunction)
if p, then q	(conditional)
if and only if	(biconditional)

Example: Classify each proposition as simple or compound. If compound, classify as negation, conjunction, disjunction, conditional or biconditional.

1. He is confidently beautiful with a heart.

2. I play Among Us, but I want to play it with my friends.
3. If you are a member of our group, then you can avail 25% discount
4. John Rey Tiangco is our congressman, but Toby Tiangco is our mayor.
5. Either you help your mother, or she will die due to over fatigue.
6. The graph of a function is a straight line if and only if its equation is in first degree.

Answer:
1. Simple proposition
2. Compound proposition; conjunction (because of the connector "but")
3. Compound proposition; conditional (if then statement)
4. Compound proposition; conjunction (because of the connector "but")
5. Compound proposition; disjunction (because of the connector "or")
6. Compound proposition; biconditional (because of the connector "if and only if")

SYMBOLS

Symbols are used to simplify work in logic. If x and y are the frequently used letters in algebra, the letters p, q, or r are often used to represent propositions in logic. The table below shows the several symbols for connectives, together with the respective types of compound proposition.

CONNECTIVE	SYMBOL	TYPE OF STATEMENT
and	$\wedge$	conjunction
or	$\vee$	disjunction
not	$\sim$	negation
if…then	$\rightarrow$	conditional
if and only if	$\leftrightarrow$	biconditional

Example: Let p represents the proposition "Ken is a hardworking person." q represents the proposition "He can accomplish many things."
Convert each compound proposition into symbols.
1. Ken is a hardworking person, and he can accomplish many things.
2. Ken is not a hardworking person, and he cannot accomplish many things.
3. It is not the case that Ken is a hardworking person or he can accomplish many things.
4. If Ken is a hardworking person, then he can accomplish many things.
5. Ken is a hardworking person if and only if he can accomplish many things.

Answer:
1. $p \wedge q$
2. $\sim p \vee \sim q$
3. $\sim(p \vee q)$
4. $p \rightarrow q$
5. $p \leftrightarrow q$

Different Types of Operations on Proposition

Let's take the statement, "*Lapu-Lapu is a brave man.*"

This statement, which we can represent with the variable ***p***, is either true or false.

p = Lapu-Lapu is a brave man.

If Lapu-Lapu is really a brave man, then ***p*** is true. If it is not, then ***p*** is false.

NEGATION

The **negation** of a statement called ***not p***, is the statement that contradicts ***p*** and has the opposite truth value.

not p = Lapu-Lapu is not a brave man.

Below are how both possibilities are represented in truth table in which *T* represents true and *F* represents false:

NEGATION	
p	$\sim p$
T	F
F	T

CONJUNCTION

A **conjunction** is a compound statement representing the word "and". For example, we have the following statements:

p = Lapu-Lapu is a brave man.

q = He fights for his countrymen's freedom.

The conjunction p and q are:

"Lapu-Lapu is a brave man, and he fights for his countrymen's freedom" This statement will only be true if both p and q are true; that is, if Lapu-Lapu is a brave man, and he fights for his countrymen's freedom. If either p or q is false, then the conjunction is false.

Here is the truth value showing the possibilities of a conjunction:

CONJUNCTION		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

DISJUNCTION

A **disjunction** is a compound statement representing the word "or". In order for the disjunction to true, one or both of the original statements has to be true. For example, we have the following statements:

p = Lapu-Lapu is a brave man.

q = He fights for his countrymen's freedom.

The disjunction p or q is:

"Lapu-Lapu is a brave man, or he fights for his countrymen's freedom." This statement is true if p or q or both statements are true; If both statements are false, then the disjunction is false.

Here is the truth value showing the possibilities of a disjunction:

DISJUNCTION		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

IMPLICATION

An **implication** is a conditional "if-then" statement like "If Lapu-Lapu is a brave man, then he fights for his countrymen's freedom."

The statement is being divided into two where the first part or the "if part" is the antecedent or called as the hypothesis and the second or the "then part" is the consequent or called as the conclusion. For example, we have the following statements:

$\boldsymbol{p}$ = Lapu-Lapu is a brave man.

$\boldsymbol{q}$ = He fights for his countrymen's freedom.

The implication "If Lapu-Lapu is a brave man, then he fights for his countrymen's freedom", will become false when the hypothesis is true, and the conclusion is false. In other words, the statement "If Lapu-Lapu is a brave man, then he will not fight for his countrymen's freedom", is false because the promise of the implication was broken.

However, if it is not a sunny day (p is false), then the promise of the implication cannot be broken since the first part (the "if part") never happened, so the implication holds true. The implication does not say what happens if Lapu-Lapu is a brave man.

Here is the truth value showing the possibilities of an implication where $p \rightarrow q$ is read as "p implies q"

IMPLICATION		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

BICONDITIONAL

The **biconditional** of a propositions or statements p and q is denoted by:

$p \leftrightarrow q$: (p if and only if q)

and is defined though its truth table:

BICONDITIONAL		
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The proposition may also be written as " p iff q". The propositions p and q are the components of the biconditional.
For example, suppose that
$\boldsymbol{p}$ = Lapu-Lapu is a brave man.
$\boldsymbol{q}$ = He fights for his countrymen's freedom.
Applying the biconditional $p \leftrightarrow q$: (p if and only if q) of the two propositions, we will have:
Lapu-Lapu is a brave man if and only if he fights for his countrymen's freedom.

What's More

"Proposition or Not"

Directions: Determine whether the following statements are propositions. Write P if it illustrates a proposition, otherwise NP.

1. Does she give you motivation to work hard?
2. If x -5 = 8, then x = 13.
3. 4 > 7
4. Keep your eyes open.
5. The 45-degree angle is a kind of acute angle.
6. Can you help me with that?
7. Wait a minute!
8. The equation x + 4 = y will give infinitely many solutions.
9. Cigarette smoking is dangerous to your health.
10. Turn right.

What I Can Do

Part 1: "Simple, Compound or Neither"
Directions: Classify each proposition as simple, compound or neither.

1. On Monday is Bonifacio Day.
2. Raise your right hand when reciting the "Panatang Makabayan".
3. Eldrian is a kind person.
4. I paid the right amount to the cashier to save time.
5. Don't cry.
6. Roger is hunting Chang'e.

Part 2: "Converting Propositions into Symbols"
Directions: Convert each compound proposition into symbols.

Given that:
P = The measure of angle A is 25 degrees
Q = Angle A is an acute angle.

7. If the measure of the Angle A is 25 degrees, then it is an acute angle.
8. The measure of the Angle A is 25 degrees, or it is an acute angle.
9. The measure of the Angle A is not 25 degrees.
10. Angle A is not an acute angle.

MODULE 8

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Propositional Logic. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two competencies, namely:

- Determining the truth values of propositions
- Illustrating the different forms of conditional propositions.

After going through this module, you are expected to:

1. Determine the truth values of propositions.
2. Illustrate the different forms of conditional propositions.

Lesson 8	Propositional Logic

What is It

LOGICAL EQUIVALENCE

The two propositions p and q are logically equivalent, denoted by $p \Leftrightarrow q$, if they have the same truth values for all possible truth values of their simple components.

Logical equivalence can also be expressed in terms of a biconditional statement. Two propositions p and q are logically equivalent if the propositions $p \leftrightarrow q$ is always true (or is a tautology).

Example:

Show that $(p \rightarrow q) \Leftrightarrow [(\sim p) \vee q]$. We shall call this logical equivalence or the Switcheroo Law.

Solution: We need to show that $p \rightarrow q$ and $(\sim p) \vee q$ have the same values using a truth table.

Step 1: Construct the two-column table for all the possibilities for p and q.

p	q
T	T
T	F
F	T
F	F

Step 2: Create another column for $p \rightarrow q$ and its values.

p	q	$p \rightarrow q$
T	T	T
T	F	T
F	T	F
F	F	T

Step 3: Create another column for ~p. To get its value, refer to the first column (p column) and apply the negation law.

p	q	$p \rightarrow q$	$\sim p$
T	T	T	F
T	F	T	F
F	T	F	T
F	F	T	T

Step 4: Create the last column for $(\sim p) \vee q$. To get its value, refer to the fourth column (~p column) and second columns then apply the disjunction.

p	q	$p \rightarrow q$	$\sim p$	$(\sim p) \vee q$
T	T	**T**	F	**T**
T	F	**T**	F	**F**
F	T	**F**	T	**F**
F	F	**T**	T	**T**

Notice that the third and fifth column of the truth table above contains the same truth values in the same sequence.

p	q	$p \to q$	$\sim p$	$(\sim p) \vee q$
T	T	**T**	F	**T**
T	F	**T**	F	**F**
F	T	**F**	T	**F**
F	F	**T**	T	**T**

Therefore $p \to q$ and $(\sim p) \vee q$ are logically equivalent. In symbol, $(p \to q) \Leftrightarrow [(\sim p) \vee q]$.

DIFFERENT FORMS OF CONDITIONAL PROPOSITIONS

Suppose p and q are propositions.

From the conditional proposition $p \to q$, we derive three other conditional statements, namely its

Converse: $q \to p$
Inverse: $\sim p \to \sim q$
Contrapositive: $\sim q \to \sim p$

Note that a conditional and the corresponding contrapositive are logically equivalent. Likewise, for a given conditional statement, its converse and inverse are logically equivalent.

Example 1: Consider that the conditional statement below is true.

$p \to q$: If Mathilda is in Grade 11, then she is a senior high school student.

State its:

1. Converse: $q \to p$
2. Inverse: $\sim p \to \sim q$
3. Contrapositive: $\sim q \to \sim p$

Solution:

1. Converse ($q \to p$): If Mathilda is a senior high school student, then she is in Grade 11.

2. Inverse ($\sim p \to \sim q$): If Mathilda is not a Grade 11, then she is not a senior high school student.

3. Contrapositive ($\sim q \to \sim p$): If Mathilda is not a senior high school student, then she is not in Grade 11.

Example 2:

Let $p \to q$: If 3x + 5 = 26, then x = 7.

State its:

1. Converse: $q \to p$

2. Inverse: $\sim p \to \sim q$

3. Contrapositive: $\sim q \to \sim p$

Solution:

1. Converse $(q \to p)$: If x = 7, then 3x + 5 = 26.

2. Inverse $(\sim p \to \sim q)$: If 3x + 25 ≠ 26, then x ≠ 7.

3. Contrapositive $(\sim q \to \sim p)$: If x ≠ 7, then 3x + 5≠26.

The Negation of a Conditional Statement

The negation of $p \to q$ is $p \wedge \sim q$. This can be expressed as $\sim(p \to q) \Leftrightarrow p \wedge \sim q$.

p	q	$p \to q$	$\sim q$	$p \wedge \sim q$	$\sim(p \to q)$
T	T	T	F	**F**	**F**
T	F	F	T	**T**	**T**
F	T	T	F	**F**	**F**
F	F	T	T	**F**	**F**

The truth table shows that $p \wedge \sim q$ negates $p \to q$ since the two columns have opposite truth values.

Example: Write the negation of the following:

1. If there is unwanted noise, then I cannot concentrate on my studies.
2. $\sim M \to \sim N$

Solution:

1. If there is unwanted noise, then I cannot concentrate on my studies.

Copy ↓ Change if-then to and Negate ↓

If there is no unwanted noise, then I can concentrate on my studies.
Answer: If there is no unwanted noise, then I can concentrate on my studies.

2. Given: $\sim M$ $\to$ $\sim N$

Copy ↓ Change if-then to and Negate ↓

$\sim M$ $\wedge$ N

Answer: $\sim M \wedge N$

What's More

"Filling Out the Truth Tables"

Directions: Fill out the following truth tables.

1.

$p \wedge \sim p$		
p	$\sim p$	$p \wedge \sim p$
T		
F		

2.

$p \wedge \sim q$			
p	q	$\sim q$	$p \wedge \sim q$
T	T		
T	F		
F	T		
F	F		

What I Can Do

"Constructing Truth Tables"

Directions: There are given statements below. Construct a truth table for each and highlight its truth values.

1. $\sim(Q \wedge R)$
2. $\sim(\sim Q \wedge \sim R)$
3. $P \wedge \sim R$
4. $\sim P \vee \sim R$
5. $R \vee (\sim P \wedge R)$
6. $P \vee R \rightarrow \sim R$
7. $\sim P \wedge \sim Q \leftrightarrow \sim(P \wedge R)$
8. $P \leftrightarrow Q \wedge R$
9. $(P \vee \sim Q) \leftrightarrow (P \rightarrow Q)$
10. $(P \rightarrow \sim Q) \vee (\sim P \vee Q)$

Assessment

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

1. What do you call the interest that is computed on the principal and then added to it?

   A. Simple Interest
   B. Compound Interest
   C. Loan Date
   D. Maturity value

2. What do you call the interest computed on the principal and on the accumulated past interest?

   A. Simple Interest
   B. Compound Interest
   C. Maturity Value
   D. Loan Date

3. At what simple interest rate per annum will ₱30,000 accumulate to ₱51,000 in 10 years?

   A. 8% B. 5% C. 7% D. 6%

4. How much interest is charged when ₱100,000 is borrowed for 10 years at an annual interest rate of 12%?

   A. ₱1,200 B. ₱120 C. ₱1,250 D. ₱200

5. It is a series of equal payments at regular intervals.

   A. Annuity
   B. Bond
   C. Stocks
   D. None of these choices

6. Vinz deposits Php 36,000 in a bank which pays interest at 9%, compounded semi-annually. He plans to withdraw Php 4,500 every 6 months. The first of which is to be made at the end of 4 years. How long will his investment last?

   A. 3 years
   B. 4 years
   C. 8 years
   D. 9 years

7. The per-share amount, as stated on the company certificate. It is determined by the company and remains stable over time.

   A. Par Value
   B. Stock Market
   C. Market Value
   D. Common Stocks

8. It is a money lent to an individual for personal or family purposes.

   A. Business loan
   B. Consumer loan
   C. both A & B
   D. none of these choices

9. Which among the logical connectors must be present in a compound proposition?

   A. On
   B. out
   C. if
   D. but

10. "If Cecilion is a vampire, then he eats blood." State its inverse.

   A. If Cecilion eats blood, then he is a vampire.
   B. If Cecilion doesn't eat blood, then he is a vampire.
   C. If Cecilion is not a vampire, then he doesn't eat blood.
   D. If Cecilion doesn't eat blood, then he is not a vampire.

What's More	What's I can Do?
1. The first payment is at time 7. The period of deferral will be from time 0 to 6, which is equivalent to **6 periods or 6 months**. 2. Five years from now is at time 5. The period of deferral will be from time 0 to time 4. Thus, the period of deferral is **4 periods or 4 years**. 3. Two years from now will be at time 8 if one quarter is considered as one period. Thus, the period of deferral is from time 0 to time 7, which is equivalent to **7 quarters or 7 periods**. **4.** The first payment is due five years from now which is equivalent to time 10 if payments are made semi-annually. The period of deferral will be from time 0 to time 9, which is equivalent **to 9 semi-annual intervals or 9 periods.** **5.** The first payment is due at the end of 6 years which is at time 3 if payments are made every 2 years. The period of deferral is from time 0 to time 2, which is equivalent **to 2 periods or 2 two-year intervals.**	1. P 107,088.20 2. P 39,441.14 3. P 1,089,533.99 4. P 57,922.41 5. P 32,346.70

Module 4

What Can Do	What's More
1. B 2. C 3. C 4. C 5. D 6. D 7. C 8. A 9. A 10. C	A. 1. 26,318.76 or 26, 320 2. 47,566.14 or 47,565 3. 340,425.50 or 340,400 4. 15,134.23 or 15,159.04 5. 221,444.20 or 221,666.50 B. 1. 1,362,245.01 or 1,429,586.25 2. 3,187.30 or 3,186.38 3. 19,647.71 or 19,638.41 4. 23,312.78 or 23,309.94 5. 9,791.26 or 9,791.23

Module 3

What I Know
1. A 2. B 3. A 4. A 5. A 6. B 7. D 8. A 9. B 10. B

Answer Key

Assessment

1. A
2. B
3. C
4. A
5. A
6. A
7. A
8. B
9. D
10. C

What's I can Do?

1. $F = (1 + r)^t = 100{,}000(1 + 0.08)^3 = 251{,}942.40$
2. P3,040,000
3. P1,275,000
4. $\boldsymbol{P_n = A\left[\frac{1-(1+i)^{-n}}{i}\right]} = 183{,}026.37\left[\frac{1-(1+0.06)^{-5}}{0.06}\right] = 770{,}973.65$
5. $\boldsymbol{A} = \frac{\boldsymbol{P_n}}{\left[\frac{1-(1+i)^{-n}}{i}\right]} = A = \frac{30{,}000}{\left[\frac{1-(1+0.0075)^{-12}}{0.0075}\right]} = 2,623.54$

What's More

1. Business loan
2. Business loan
3. Consumer loan
4. Consumer loan
5. Consumer loan

References

Module 1

Verzosa et al. (2016). *Senior High School General Mathematics Teacher's Guide SHS v. 1*, Commission on Higher Education, pp. 159-176.

Module 2

Verzosa, et.al, (2016). *Teaching Guide for Senior High School General Mathematics* pp. 159 – 176, Commission on Higher Education.

Module 3

DepEd-BLR, First Edition. (2016). *General Mathematics Teacher's Guide SHS v. 1*, DepEd Central Office pp. 188-212.

Oronce, O. A. (2016). *General Mathematics*. Rex Book Store, Inc. pp. 203-242

Module 4

DepEd-BLR, First Edition. (2016). *General Mathematics Teacher's Guide SHS v. 1*, DepEd Central Office pp. 213-221.

Oronce, O. A. (2016). *General Mathematics*. Rex Book Store, Inc. pp. 224-227

Module 5

Versoza, DM, B., Apolinario, P, L., et al. (2016). *General Mathematics*. p 210.

PESOLAB, A. (2020, September 04). Blue chips: Philippines' best 30 companies in PSEi. Retrieved November 30, 2020, from https://pesolab.com/blue-chips-philippines-best- 30-companies-in-psei/Listed companies in the Philippines

(n.d.). INVESTMENT MARKETING PORTAL. https://imlinked.prulifeuk.com.ph/UploadedFiles/FundFactSheets/sub224 1-11-24-202010-53-19-PM.pdf

STOCKS AND BONDS

Module 6

DepEd-BLR, First Edition. (2016). *General Mathematics Teacher's Guide SHS v. 1*, DepEd Central Office pp. 237-248.

Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc. pp. 203-242

Module 7

Oronce, O. A. (2016). General Mathematics. Rex Book Store, Inc. pp. 266-277.

Verzosa et al. (2016). *Senior High School General Mathematics Teacher's Guide SHS v. 1*, Commission on Higher Education, pp. 159-176.

Module 8

DepEd-BLR, First Edition. (2016). *General Mathematics Teacher's Guide SHS v. 1*, DepEd Central Office pp. 188-212.

Oronce, O. A. (2016). *General Mathematics*. Rex Book Store, Inc. pp. 203-242

For inquiries or feedback, please write or call:

Department of Education – Schools Division Office Navotas
Learning Resource Management Section

Bagumbayan Elementary School Compound
M, Naval St., Sipac Almacen, Navotas City

Telefax: 02-8332-77-64
Email Address: navotas.city@deped.gov.ph